

Development of a 3D ray tracing-based direct solar shading model for urban building energy simulation

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ARTICLE INFO

Keywords:

Solar
Shading
Ray tracing
UBEM
Building energy

ABSTRACT

Direct solar shading by nearby buildings plays a significant role in urban building energy models (UBEMs) for predicting building energy performance. Analytical methods currently dominate solar shading analysis in UBEMs; however, (I) they require geometry simplifications to process complex geometries, such as buildings with curved surfaces, and struggle to model structures like trees, and (II) they rely on hard-to-replicate computational acceleration schemes for large-scale simulations due to repeated surface-by-surface shading analysis. These shortcomings highlight the need to explore alternative approaches for direct solar shading simulations required by UBEMs. In this paper, we developed a ray tracing-based solar shading model designed for large-scale simulations, which can successfully process complex structures. Our 3D shading model tracks sun rays across the urban area simultaneously, eliminating repeated shading calculations and, thereby, the need for traditional acceleration schemes necessary for analytical methods. The model's accuracy is demonstrated through comparisons with measurements and results from existing tools. Furthermore, the model is integrated into our in-house UBEM, CityBEM, to enhance solar radiation assessments. Simulation results for a downtown Montréal (Quebec, Canada) test case suggest that neglecting solar shading led to an average underestimation of winter energy use by 13% and an average overestimation of summer energy use by 15%. Moreover, rooftop solar radiation was overestimated by more than 20% for over a third of buildings, while about 30% of buildings had overestimations of 10%–30%. Future work will incorporate additional factors, such as geometry partitioning, GPU acceleration, and adaptive model inputs, for scalable urban simulations. In conclusion, our research provides a valuable tool for improving UBEM predictions and supporting sustainable urban design.

Nomenclature

Symbols	
A	Area (m^2)
A_i	Anisotropy index (–)
ACH	Air change per hour (h^{-1})
a	Constant coefficient (–)
b	Constant coefficient (–)
C	Thermal capacity ($\text{J}\cdot\text{K}^{-1}$)
C_p	Specific heat ($\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$)
COP	Coefficient of performance (–)
c	Constant coefficient (–)

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Symbols	
d	Constant coefficient (m)
E	Longwave radiation ($\text{W}\cdot\text{m}^{-2}$)
F_{inf}	Air infiltration rate ($\text{m}^3\cdot\text{s}^{-1}$)
$f_{shading}$	Direct solar shading factor (–)
h	Convection heat transfer coefficient ($\text{W}\cdot\text{m}^{-2}\cdot\text{K}^{-1}$)
I_{bn}	Beam radiation at normal incidence ($\text{W}\cdot\text{m}^{-2}$)
I_{db}	Direct beam radiation ($\text{W}\cdot\text{m}^{-2}$)
I_{gd}	Ground-reflected diffuse radiation ($\text{W}\cdot\text{m}^{-2}$)
I_{on}	Extraterrestrial radiation at normal incidence ($\text{W}\cdot\text{m}^{-2}$)
I_{sd}	Sky diffuse radiation ($\text{W}\cdot\text{m}^{-2}$)

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Symbols	
I_t	Total solar radiation ($\text{W}\cdot\text{m}^{-2}$)
k	Thermal conductivity ($\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$)
L	Layer thickness (m)
$\dot{M}_{HVAC}$	HVAC system air flow rate ($\text{kg}\cdot\text{s}^{-1}$)
N	Number of elements (–)
n	Normal vector component (–)
PC	Power consumption (W)
Q	Heat gain/loss (W)
$q_{solar, glz}$	Solar energy flow through glazing ($\text{W}\cdot\text{m}^{-2}$)
R	Thermal resistance ($\text{m}^2\cdot\text{K}\cdot\text{W}^{-1}$)
R_b	Ratio of direct beam radiation on tilted surface to beam on horizontal (–)
R_d	Ratio of sky diffuse radiation on tilted surface to diffuse on horizontal (–)
R_r	Ratio of reflected radiation on tilted surface to total radiation on horizontal (–)
S	Source term representing a heat source ($\text{W}\cdot\text{m}^{-2}$)
$SHGC$	Solar heat gain coefficient (–)
T	Temperature (K)
t	Time (s)
τ_{ray}	Distance along a ray (m)
V	Volume (m^3)
x	Spatial coordinate (m)
y	Spatial coordinate (m)
z	Spatial coordinate (m)

Abbreviations

UBEM Urban building energy model

Greek letters

α	Solar shortwave absorptance (–)
β	Tilt angle (degree)
γ	Azimuth angle (degree)
ϵ	Longwave emissivity (–)
η	Efficiency (–)
θ	Glazing thermal node temperature (K)
θ_{inc}	Angle of incidence of solar beam radiation on a tilted surface (degree)
θ_z	Zenith angle (degree)
ρ	Density ($\text{kg}\cdot\text{m}^{-3}$)
ρ_g	Ground shortwave reflectance (–)

Subscripts

b	Building
c	Cooling
$conv$	Convection heat transfer
ctr	Geometric center
ctr, ig	Geometric center of inlet grids
gg	Geometry grid
glz	Glazing
h	Heating
hor	Horizontal surface
i	Spatial notation
in	Indoor
inf	Infiltration
int	Intersection point
out	Outdoor
out, bf	Outside the building footprint
rad	Longwave radiation interaction
shd	Direct solar shading impacts
srg	Solar shortwave radiation heat gain
sup	Supply air
sur	Building surface
t	Total
$th, mass$	Thermal mass
x	x-axis direction
y	y-axis direction

1. Introduction

Assessing solar radiation on building façades in urban environments is a complex task [1]. This assessment should consider several factors, including direct solar radiation (or direct beam radiation), diffuse sky radiation, reflected radiation from the ground, and multi-reflections

between building surfaces. A greater challenge arises in dense urban areas, where nearby buildings, trees, and other structures can influence the radiation reaching façades. In particular, direct solar shading cast by neighboring structures can significantly reduce solar exposure [2], affecting heating and cooling demands, energy consumption [3], and the performance of solar photovoltaic panels on-site [4]. These shading effects should be included in bottom-up physics-based urban building energy models (UBEMs) to improve the accuracy of energy predictions. Modeling direct solar shading patterns can guide sustainable building design, allowing for the optimization of factors such as shape, height, and orientation of buildings to reduce their energy consumption [5]. In the rest of this paper, we will refer to direct solar shading as “solar shading”.

1.1. Impact of solar shading on building energy performance

Solar shading can significantly impact building energy use [6]. Publications report a wide range of impacts, from less than 10% to over 50%, depending on factors such as weather conditions, building usage type, and the arrangement of surrounding structures [7]. When solar shading is neglected in building energy models, it typically results in underestimated heating energy use during winter and overestimated cooling energy use in summer [8]. Zhao et al. [9] projected average energy use underestimations of 7%–36% in heating and overestimations of 14%–48% in cooling across different climate zones in China compared to scenarios where solar shading was ignored. Liu et al. [10] investigated nine groups of buildings with different arrangements and heights across several climate zones, realizing that the cold climate zone exhibited the highest energy use overestimations (up to 45%) in summer and underestimations (up to 21%) in winter due to neglecting shading impacts. Shahidi et al. [11] examined the impact of surrounding buildings' height on a target building's energy use in a Canadian cold climate. Their study showed that neglecting solar shading led to energy use prediction differences by up to 44% in both cooling and heating.

Beyond its effect on building energy use, solar shading can also impact the performance of locally installed renewable energy systems, such as photovoltaic (PV) panels. For instance, overlooking solar shading can significantly overestimate on-site photovoltaic electricity generation, leading to unrealistic estimates of carbon emission savings from photovoltaic retrofitting [7,12]. Previous studies have explored optimizing building arrangements to reduce energy use and maximize solar potential by considering shading effects [9].

1.2. Modeling direct solar shading in urban areas

Several methods are available to model solar shading effects in building energy models, such as pixel counting, radiosity, ray tracing, and analytical approaches [13,14]. In pixel counting, light exposure is calculated by generating image buffers from the light source's perspective and counting the visible pixels on surfaces [15]. Pixel counting is less common in urban-scale simulations due to its reliance on image resolution, which can compromise the accuracy of predictions [14]. The radiosity method solves the radiant energy balance for each surface in the scene, with view factors between surface pairs determining energy contributions [16]. Although radiosity is primarily used for diffuse shading effects, it can simulate realistic lighting and shadows. Radiosity becomes computationally expensive for large-scale simulations because of calculating view factors for all surface pairs in the urban scene [17], which makes it less practical for such simulations. The Simple Radiosity Algorithm (SRA) [18] has been developed to simulate solar radiation on building surfaces in urban neighborhoods, including the impacts of solar shading. In ray tracing, light rays are projected from the light source, and their path is traced until they interact with surfaces [19]. Ray tracing is commonly used in computer graphics for realistic lighting simulations. This method can be adapted for urban-scale solar shading simulations, but its application in the urban context is limited in

the literature. While ray tracing is widely recognized as a powerful technique for daylighting analysis (especially in tools like Radiance [20]), it has not traditionally been the default method in most mainstream building energy models for direct solar shading calculations. Instead, many building energy models rely on more simplified approaches such as TRNSYS [21] or analytical approaches for solar shading analysis like EnergyPlus [22]. Ray tracing approaches are slowly becoming popular for specific applications such as urban simulations because of increased computational power and a growing interest in more accurate 3D modeling [23]. Lastly, analytical methods use coordinate transformations to project shadows onto target surfaces. Techniques like polygon clipping are then used to subtract and calculate shadow fractions [22].

Building energy models such as EnergyPlus, DeST, ESP-r, and DOE-2 commonly utilize analytical approaches for modeling solar shading [13, 24]. Analytical methods calculate surface-by-surface shading repeatedly to determine shadow areas, taking into account the effect of all shading surfaces on the target surfaces, even the ones with minimal impacts, such as distant surfaces [13]. In this context, the ‘target’ refers to the object being shaded, while ‘shading’ objects cast shadows onto the target. Nevertheless, these calculations are computationally demanding [13] and the processing time increases quadratically as the number of surfaces increases [25]. This rapid increase in computation time hinders the direct application of analytical methods to urban-scale simulations. In addition, analytical methods are incapable of processing complex structures, such as buildings with curved surfaces or holes, as well as modeling objects like trees with dense shading surfaces [22].

Previous studies have sought to adopt analytical methods for urban-scale simulations by introducing preprocessing methods and computational acceleration schemes, including:

- i. Simplifying input geometry or reducing the number of exterior surfaces [26,27]: This preprocessing method reduces complexity by simplifying building geometry. For example, a building’s footprint might be replaced with a simple polygon, or multi-story façades might be merged into a single surface. While this method improves computational efficiency, it compromises accuracy.
- ii. Filtering shading structures based on a distance threshold [13, 14]: This method excludes structures that are far away and unlikely to cast shadows on the target by applying a distance threshold. Previous studies have recommended using a threshold based on building height and local solar altitude that captures over 80% of annual solar exposure [28].
- iii. Sunlight channel filtering [13,14]: In this method, nearby structures that do not cast shadows on the target at a given sun position are excluded. As the sun’s position changes throughout the day, only nearby structures that influence the target at specific times are retained, while others are excluded. For each new position of the sun, this filtering process is repeated.
- iv. Filtering non-impacting surfaces of nearby structures [13]: This step removes surfaces of nearby shading structures that do not influence the target surface. Not all surfaces of a shading structure contribute to the shadowing effect on the target, and therefore, surfaces that do not impact the target can be excluded from the analysis. In a similar manner to the sunlit channel, this filtering process needs to be applied repeatedly for each target surface and sun position.
- v. Parallel processing methods [29,30]: As shading calculations depend only on geometry and sun position, parallel processing allows for faster calculations by performing multiple shading simulations simultaneously.
- vi. Updating shading calculations based on representative days [31]: This approach reduces computational time by conducting shading calculations for a limited set of representative days throughout the year.

In summary, analytical methods are widely used for solar shading calculations in mainstream building energy models. These methods, however, are not suitable for modeling real-world and large-scale urban geometries. To adapt analytical methods for such simulations, several preprocessing and acceleration schemes have been proposed in the literature. Even though these approaches have improved their feasibility, some challenges remain.

1.3. Challenges of modeling solar shading in UBEMs

Modeling solar radiation on building surfaces is essential for physics-based UBEMs, since solar radiation affects thermal loads, daylight availability, and on-site solar photovoltaic performance. Modeling and incorporating the effects of direct solar shading into urban simulations has remained a significant challenge in UBEMs [32,33]. There is a significant computational burden associated with shading calculations in urban environments, where thousands of surfaces may interact dynamically with sunlight. Analytical shading methods (e.g., the shading model in EnergyPlus) do not scale well to urban geometries due to repeated surface-by-surface calculations, requiring extensive preprocessing, filtering, and complex computational acceleration strategies.

While shading calculation features are available in UBEM engines such as EnergyPlus, many of them either simplify or exclude shading entirely due to the complexity of implementation and the high computational costs involved, as can be seen in the UBEM works done by Johari et al. [34], Qian Ang et al. [35], and Battini et al. [36]. As an example, Battini et al. [36] approximated solar shading by replacing portions of windows with opaque surfaces, which simplifies shading modeling, but significantly compromises physical accuracy. Some UBEMs utilize simplified solar tools to account for shading. For example, the EnergyPlus-based UBEM tools4cities [37] incorporates the Simple Radiosity Algorithm (SRA) to estimate solar shading. Several in-house UBEMs also suffer from similar limitations [38,39], as they do not yet account for solar shading due to the challenges of handling large urban geometries and implementing efficient preprocessing pipelines. As a result of these ongoing challenges, UBEM workflows require more practical and scalable solutions to integrate solar shading.

1.4. Research question and contributions of the current study

Despite previous efforts to adopt analytical methods for urban-scale simulations, modeling solar shading in urban environments has remained challenging. These previous studies have primarily relied on simplified test cases with idealized building shapes and small-scale urban neighborhoods (for example, see Wang et al. [13], Liu et al. [14], and Zhou et al. [30]), raising concerns about the applicability of analytical methods to real-world, complex, and large urban environments. While analytical methods have currently dominated shading analysis in existing UBEMs, we consider that they are not the ultimate solution for real-world, large-scale simulations, opening the discussion for alternative approaches to overcome their shortcomings. Ray tracing is becoming increasingly popular as an alternative strategy for modeling solar shading in complex urban settings. By simulating the path of sunlight across the urban scene, this method enables detailed shading analyses by capturing the interactions between solar rays and surrounding structures. Although ray tracing techniques are not new, they have been mainly applied to computer visions and/or daylighting, such as the Radiance software [20]. It can be said that implementing a 3D ray tracing approach for UBEM applications is challenging, mostly when dealing with the details of the urban layout and structures in urban areas. Other studies, such as Tian et al. [40], used digital surface model (DSM)-based geometry for ray tracing shading analysis via DAYSIM. Their methodology requires extensive geometry preprocessing and translation into Radiance-compatible formats, often involving simplifications that will reduce accuracy. Additionally, DAYSIM is primarily

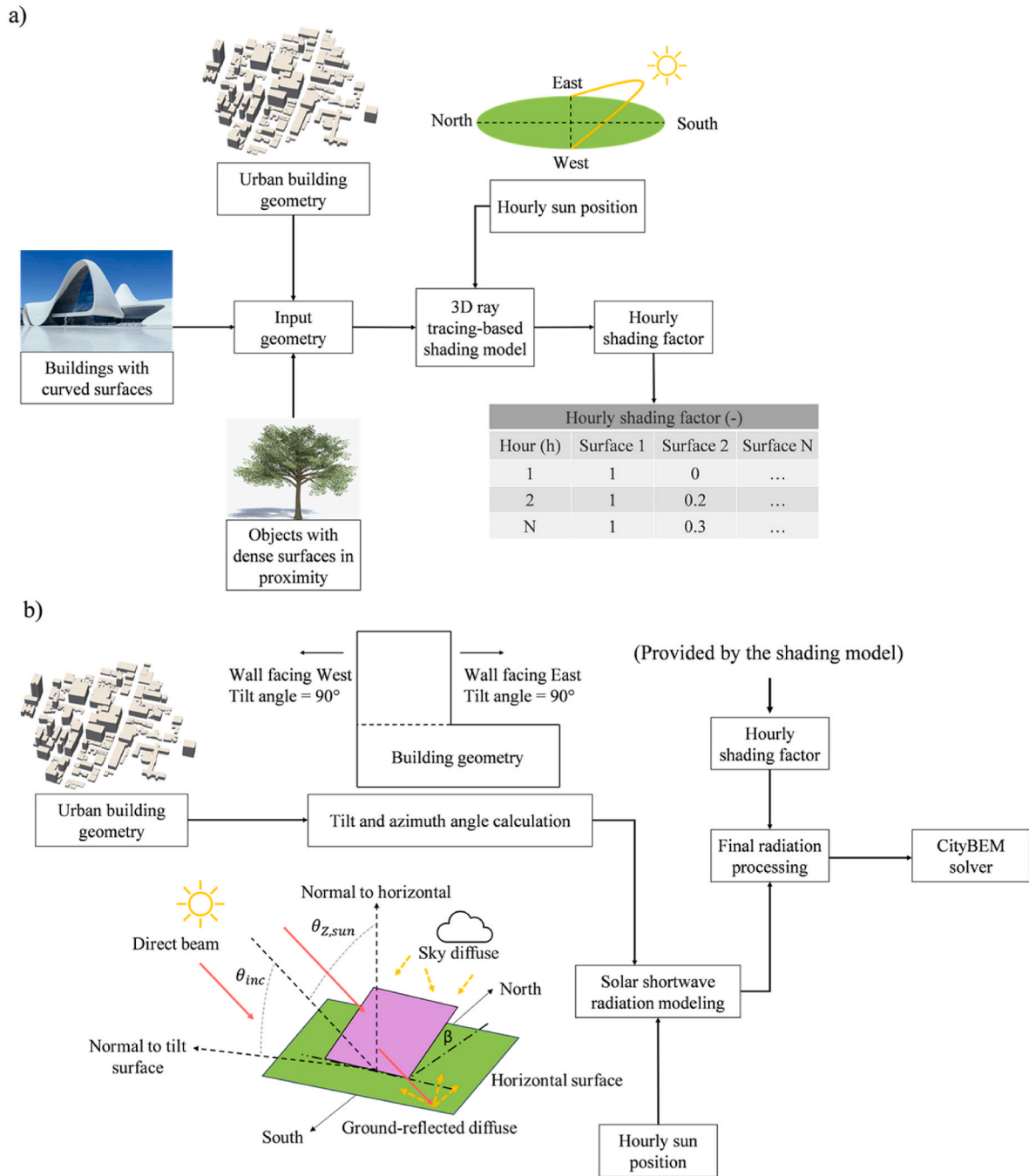


Fig. 1. Schematic of the present paper methodology: (a) general workflow of the shading model, from inputs to outputs, and (b) various steps for solar radiation modeling in CityBEM, considering shading impacts.

designed for individual buildings or small clusters of buildings and suffers from scalability for large urban areas. Other studies like Liao et al. [41] proposed a fast AI-based method for urban shading analysis; however, they only modeled rooftop shading and neglected vertical surfaces, which play an important role in assessing solar gain through glazing. It is also worth noting that AI-based models may struggle to generalize in complex urban environments without sufficiently comprehensive training data.

Thus, we proposed the following research question: Can ray tracing-based methods be adopted for urban simulations to address the challenges of analytical methods in solar shading modeling required by UBEMs?

In response, we developed a ray tracing-based model using a 3D grid

approach for solar shading simulations required by UBEMs. Our approach enables scalable, fast simulations across large urban areas with minimal computational overhead and is able to model shading from a wide variety of urban objects, including complex building geometries and trees, and integrates seamlessly with standard GIS data formats for UBEM applications. Moreover, previous shading studies have lacked a transparent integration into UBEM workflows, and their performance within complete UBEM simulations has remained unexplored [13,40]. In contrast, we further integrated the shading model into our in-house UBEM, CityBEM [42,43], to improve building energy predictions by performing a detailed assessment of solar radiation reaching building surfaces, as well as demonstrating practical applications in real-world large urban areas.

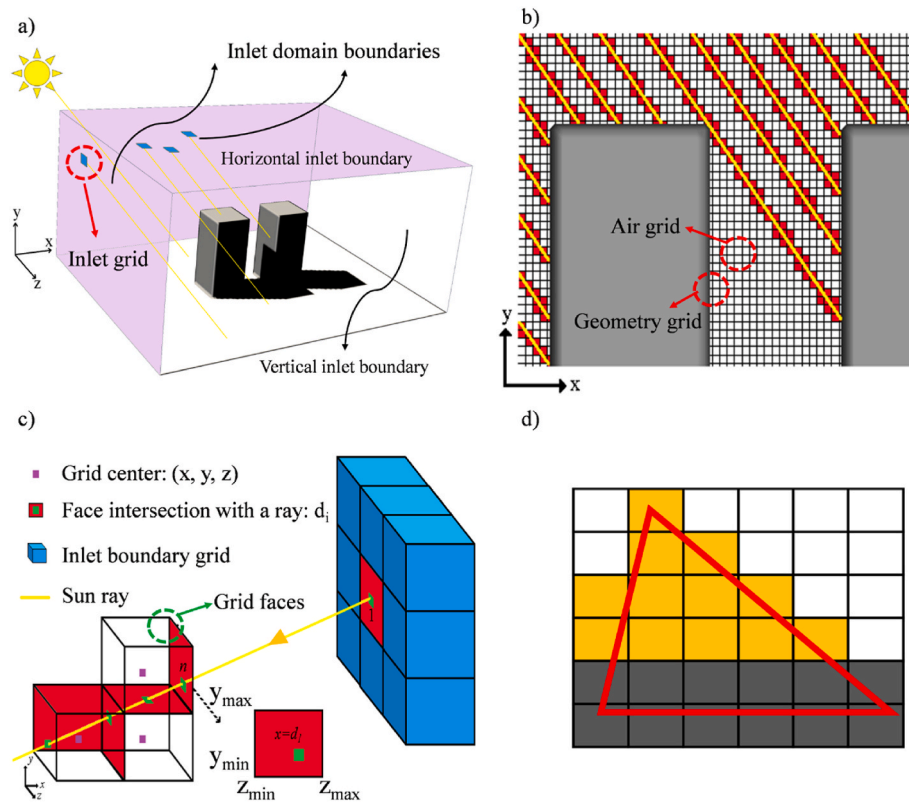


Fig. 2. Schematic of the shading model in this study: (a) horizontal and vertical inlet domain boundaries with inlet grids, (b) trajectory of the sun rays within the domain, (c) intersection of sun rays with grid faces, and (d) an exterior surface with sunlit and shaded geometry grids.

Although the proposed 3D ray tracing method is not currently implemented for parallel computing, it could be converted to GPU computing for further speedups and scalability, particularly given recent advancements in GPU parallel schemes. Overall, by addressing key gaps in solar shading analysis, this study makes a significant contribution to

urban building energy modeling, which supports future sustainable urban design efforts.

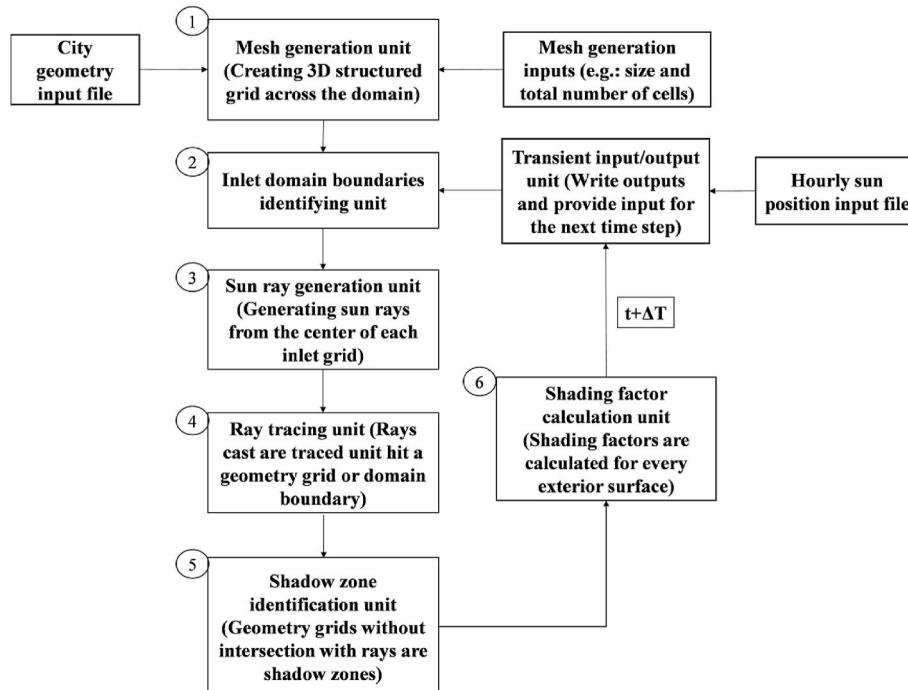


Fig. 3. Schematic of the six steps for calculating transient direct solar shading in this study.

2. Methodology

Fig. 1 shows a schematic of the current study's methodology, from developing a 3D ray tracing-based shading model to the assessment of solar radiation on building surfaces within CityBEM. As shown in Fig. 1 (a), the input geometry to the shading model can include real-world urban building geometry, complex structures with curved surfaces, or even objects like trees. The shading model reads hourly sun position data to calculate hourly shading factors on exterior surfaces. The generated shading factors indicate the fraction of exterior surfaces exposed to direct sunlight.

Fig. 1(b) illustrates various steps undertaken by CityBEM to calculate solar radiation on building surfaces. CityBEM first processes the geometry of urban buildings to determine the tilt and azimuth angles of building surfaces. These angles are essential for determining the orientation of each surface relative to the sun, which directly impacts the amount of solar radiation received. CityBEM then uses physics-based models to calculate total solar radiation, including direct beam, sky diffuse, and ground-reflected diffuse radiation. Finally, the shading factors are used to adjust the calculated solar radiation.

2.1. Development of the direct solar shading model

The shading model developed in this study is based on ray tracing and 3D grid numerical methods. It assumes that the sun is a point light source, and its distance from the Earth is significantly greater than the diameter of the Earth. Therefore, the sun's rays reaching urban areas are considered parallel.

The model first partitions the geometry region into a uniformly 3D structured grid and identifies the appropriate inlet domain boundaries for the sun rays based on the location of the sun (Fig. 2(a) and (b)). The parallel sun rays are then cast from the grids located on the inlet domain boundaries and are traced to their destination (see Fig. 2(a) and (b)). The destination of the rays can be either grids located on the exterior surfaces of structures (e.g., buildings) or the domain boundaries (without hitting exterior surfaces). Fig. 2(c) illustrates the schematic of tracing a sun ray intersecting a cluster of grids, where the green spots correspond to the points at which the ray intersects the grids' faces. Moreover, Fig. 2 (d) depicts an exterior surface (triangle) with geometric grids, in which yellow indicates the portion of the surface that receives sunlight and dark indicates the shaded portion.

As shown in Fig. 2, we have defined three types of grids: the inlet grid, the geometry grid, and the air grid. The inlet grids are located on the domain inlet boundaries where the sun's rays are cast (Fig. 2(a) and (c)). Geometry grids are located on the exterior surfaces of structures, while sun rays travel within air grids and stop when they hit geometry grids or domain boundaries (Fig. 2(a) and (b)).

Fig. 3 shows six main steps to calculate the shading factor at each time step, corresponding to a new sun position.

The six steps shown in Fig. 3 are discussed in more detail below:

- i. Mesh generation unit: In this step, a structured 3D grid consisting of x, y, and z points is generated across the urban domain. The exterior surfaces of structures, such as buildings, trees, and terrain, are mapped onto this computational grid. Subsequently, the grid points located on the exterior surfaces (i.e., boundary grid points) are identified. Due to the use of a structured grid, all grid cells have the same resolution and volume in the x, y, and z directions. Thus, inlet grids, air grids, and boundary grids are uniform in size and dimensions.
- ii. Inlet domain boundaries identifying unit: This step involves identifying the inlet boundaries for sun rays, determined by the sun's position. The horizontal boundary at the top of the domain consistently serves as the inlet boundary (see Fig. 2(a)). In addition, the vertical inlet boundaries are determined at each time step according to the solar azimuth angle, γ_{sun} (degree), as shown in Equation (1):

$$\begin{cases} \gamma_{sun} \in [0, 180^\circ] \rightarrow \text{East boundary} \\ \gamma_{sun} \in [180^\circ, 360^\circ] \rightarrow \text{West boundary} \end{cases} \quad \text{Eq. 1}$$

$$\begin{cases} \gamma_{sun} \in [-90^\circ, 90^\circ] \rightarrow \text{North boundary} \\ \gamma_{sun} \in [90^\circ, 270^\circ] \rightarrow \text{South boundary} \end{cases}$$

It should be noted that while the solar zenith angle is also used in sun tracking, it does not define the vertical boundaries.

- iii. Sun ray generation unit: In this unit, arrays representing the sun rays are defined for individual inlet grids. The trajectory of the sun's rays (x, y, and z coordinates) from the center of the inlet grids is mathematically described by Equations (2)–(4) [44]:

$$x(t) = x_{ctr,ig} + t_{ray} \cdot \sin \theta_{z,sun} \cdot \sin \gamma_{sun} \quad \text{Eq. 2}$$

$$y(t) = y_{ctr,ig} + t_{ray} \cdot \cos \theta_{z,sun} \quad \text{Eq. 3}$$

$$z(t) = z_{ctr,ig} - t_{ray} \cdot \sin \theta_{z,sun} \cdot \cos \gamma_{sun} \quad \text{Eq. 4}$$

where t_{ray} (m) shows the distance along the rays from the center of inlet grids, x (m), y (m), and z (m) represent the position of a particular point along the rays and $\theta_{z,sun}$ (degree) is the zenith angle of the sun. Moreover, $x_{ctr,ig}$, $y_{ctr,ig}$, and $z_{ctr,ig}$ are the coordinates of the center of inlet grids. The above three equations can be represented by Equation (5):

$$\frac{x - x_{ctr,ig}}{\sin \theta_{z,sun} \cdot \sin \gamma_{sun}} = \frac{y - y_{ctr,ig}}{\cos \theta_{z,sun}} = \frac{z - z_{ctr,ig}}{-\sin \theta_{z,sun} \cdot \cos \gamma_{sun}} = t_{ray} \quad \text{Eq. 5}$$

- iv Ray tracing unit: In this step, the sun rays are traced to find their intersection with the faces of grids. Each grid face can be defined by a general form of the surface equation, as shown in Equation (6):

$$ax + by + cz = d \quad \text{Eq. 6}$$

where a (–), b (–), c (–), and d (m) are constant coefficients. Because of using a structured grid and the Cartesian coordination system, Equation (6) can be simplified to Equation (7):

$$\begin{aligned} x &= d_1 \\ y &= d_2 \\ z &= d_3 \end{aligned} \quad \text{Eq. 7}$$

In Equation (7), for instance, d_1 represents the x-value on the grid face parallel to the $y - z$ plane. To find the intersection of a sun ray with a grid face, the fixed coordinate (e.g., $x = d_1$) is substituted into the sun ray's parametric equation (Equation (5)) to solve for t_{ray} . The parameter t_{ray} is then used to calculate the two other coordinates ($y(t)$ and $z(t)$). The intersection point can be confirmed by checking whether it lies between the minimum and maximum vertices of the faces and within the bounds of the grid. This process is repeated for all six faces of each grid, identifying intersections with the sun ray (green spots in Fig. 2(c)). The subsequent grid through which the sun ray passes can also be determined by finding the intersected face.

- v. Identifying shadow zones: Each face of a geometry grid in contact with exterior surfaces that does not intersect with sun rays is designated a shaded face. Consequently, the corresponding geometry grid is labeled as a shaded geometry grid.
- vi. Shading factor calculation unit: The shading factor for each surface is calculated by dividing the number of sunlit geometry grids ($N_{sunlit,gg}$) by the total number of geometry grids in contact with that surface ($N_{t,gg}$). Therefore, the shading factor, $f_{shading}$ (–), given a new sun position can be determined using Equation (8):

$$f_{shading} = \frac{N_{sunlit,gg}}{N_{t,gg}} \quad \text{Eq. 8}$$

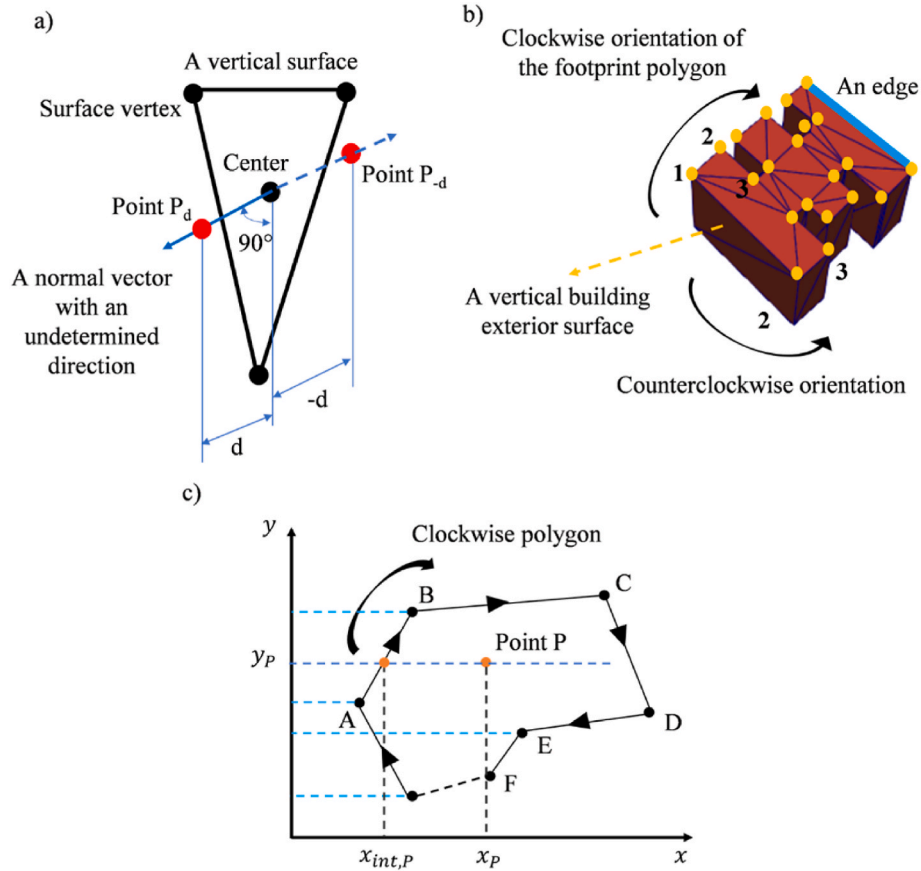


Fig. 4. a) A vertical exterior surface with an undetermined direction of the normal vector; b) Corner points forming a 2D building footprint polygon; c) Position of point P relative to the edges of a 2D polygon.

The shading factor ($f_{shading}$) is a number between 0 (fully shaded) and 1 (fully sunlit). Any shading factor between 0 and 1 measures the fraction of exterior surfaces exposed to direct sunlight. For instance, in Fig. 2 (d), the number of sunlit geometry grids ($N_{sunlit,gg}$) on the surface is 12, and the total number of geometry grids ($N_{t,gg}$) is 24. Therefore, the shading factor for this surface is 0.5.

It is also worth noting that the proposed shading methodology could be further enhanced by incorporating digital surface models (DSM) derived from LiDAR data to capture finer details of building surfaces and surrounding urban elements [45]. This approach would enable more detailed simulations and improve the representation of shading effects, particularly in cases where conventional city geometries oversimplify complex structures such as trees. However, integrating DSM data into our 3D ray tracing framework presents some challenges. Our method uses a structured 3D mesh that explicitly defines volumetric space, which enables the tracking of sun rays through air volumes until they interact with geometry boundaries. In contrast, DSMs are 2.5D surface representations that would require substantial preprocessing before they could be used with grid-based ray tracing approaches. We believe that incorporating DSM data can enhance the scalability of urban simulations.

2.2. CityBEM mathematical model and workflow

CityBEM is an in-house, bottom-up, physics-based urban building energy model developed in C++ [42]. It is designed specifically to perform high spatiotemporal resolution simulations of urban buildings, down to the minute level. Using efficient computational algorithms, CityBEM models the dominant heat transfer processes in buildings to enable fast simulations. Unlike building energy models such as

EnergyPlus or TRNSYS, which are primarily developed for detailed simulations of individual buildings, CityBEM is purpose-built for urban-scale applications and does not require extensive adaptation for UBEEM use. Currently, CityBEM is not open and available.

2.2.1. Building surface azimuth angle calculation

To predict the solar radiation incident on building surfaces, it is essential to calculate their tilt and azimuth angles. Calculating the tilt angle of surfaces is straightforward, but determining their azimuth angle can be challenging if the normal vector data in the input geometry is missing or incorrect. While the coordinates of surface vertices can be used to obtain normal vectors, additional analysis is needed to determine their correct direction (inward or outward). CityBEM currently supports modeling geometry at Level of Detail (LOD) 1, which represents buildings with flat roofs and average heights [43]. In LOD1, all building surfaces are either vertical or horizontal. CityBEM utilizes the methodology presented here to calculate the azimuth angle of building exterior surfaces.

Fig. 4(a) illustrates a vertical surface with an initially undetermined normal vector direction. Two points (P_d and P_{-d}) are defined at distances d and $-d$ from the surface center to help identify the correct direction of the normal vector. In fact, the point outside the building footprint indicates the proper direction of the normal vector. To facilitate further analysis, a 2D polygon representing the building's footprint is constructed using ordered corner points, as shown in Fig. 4(b).

The relative positions of P_d and P_{-d} to the edges of the footprint polygon are examined to determine their status (i.e., whether located outside the building footprint). To do so, P_d and P_{-d} should be projected onto the horizontal plane to represent 2D points. Subsequently, if a horizontal line passing through point P ($y = y_P$) intersects any edges of

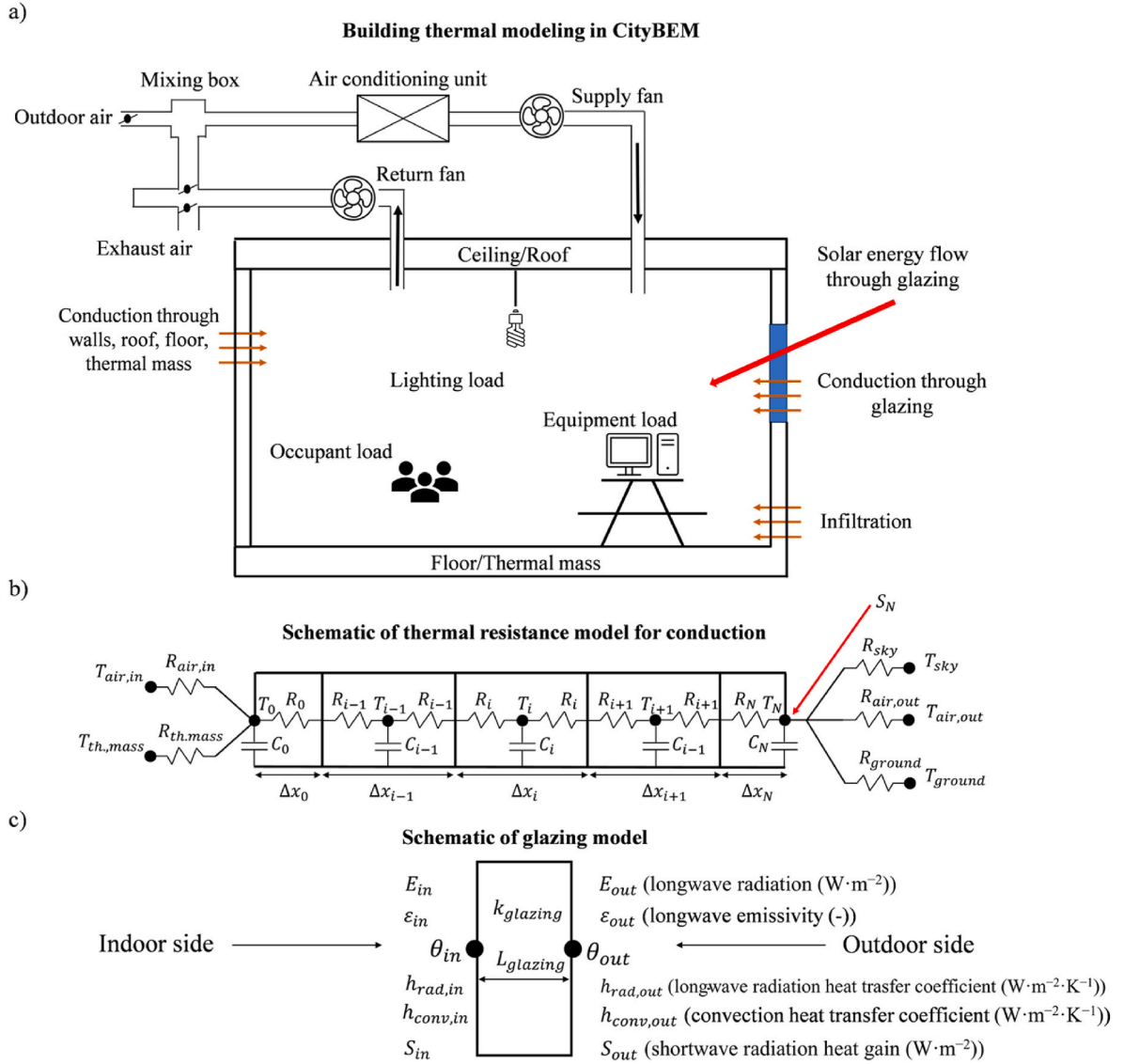


Fig. 5. Building thermal modeling in CityBEM [42,43]: (a) Schematic of single thermal zone modeling; (b) Thermal resistance model for conduction; (c) Various terms considered in glazing modeling.

the polygon and lies to the right of these edges, it indicates that point P is inside the polygon, assuming the orientation is clockwise. According to Fig. 4(c), point P is inside the polygon since it is located on the right side of edges AB and CD while the orientation is clockwise.

The process mentioned above can be translated into the following steps:

- i. Intersection check: If the y-coordinate of point P (y_p) lies between the y-coordinates of a polygon edge, the horizontal line passing through P will intersect that edge.
- ii. Inside/outside the polygon check: For this purpose, the x-coordinate of point P (x_p) is compared with the x-coordinate intersection of $y = y_p$ with polygon edges ($x_{int,p}$). Point P is inside the polygon if $x_p > x_{int,p}$ and the orientation is clockwise, and vice versa. The subscript *int* shows the intersection point. The x-coordinate intersection of $y = y_p$ with the edge AB can be obtained using Equation (9):

$$x_{int,p} = \frac{x_B - x_A}{y_B - y_A} \cdot (y_p - y_A) + x_A \quad \text{Eq. 9}$$

Following the identification of the correct direction, the first two components of the normal vector (n_x and n_y) for the corresponding

building surface can be determined using Equations (10) and (11):

$$n_x = x_{P,out,bf} - x_{sur,ctr} \quad \text{Eq. 10}$$

$$n_y = y_{P,out,bf} - y_{sur,ctr} \quad \text{Eq. 11}$$

where subscript *P, out, bf* represents the point among P_d or P_{-d} located outside the building footprint. The corresponding surface azimuth angle, γ_{sur} (degree), is then calculated using Equation (12):

$$\gamma_{sur} = \text{atan2}(n_y, n_x) \cdot \frac{180^\circ}{\pi} \quad \text{Eq. 12}$$

To ensure positive azimuth angles, the adjustment given in Equation (13) is applied:

$$\gamma_{sur} = \gamma_{sur} + 360^\circ \text{ if } \gamma_{sur} < 0 \quad \text{Eq. 13}$$

2.2.2. Calculation of solar radiation on building surfaces

Solar radiation incident on tilted surfaces includes direct beam, sky diffuse, and ground-reflected diffuse radiation. CityBEM uses the equations presented here to calculate the solar radiation incident on building surfaces.

The direct beam radiation on a tilted surface is determined using the geometric factor R_b (–), which is defined in Equation (14) [21]:

$$R_b = \frac{\cos \theta_{inc}}{\cos \theta_{z, sun}} \quad \text{Eq. 14}$$

where θ_{inc} (degree) is the angle of incidence of direct beam radiation on a tilted surface. The cosine of θ_{inc} can be calculated using Equation (15) [21]:

$$\cos \theta_{inc} = \cos \theta_{z, sun} \cdot \cos \beta_{sur} + (\sin \theta_{z, sun} \cdot \cos (\gamma_{sun} - \gamma_{sur}) \cdot \sin \beta_{sur}) \quad \text{Eq. 15}$$

Here, β_{sur} (degree) is the building surface tilt angle, and γ_{sur} (degree) is the building surface azimuth angle, obtained from Equation (12) and Equation (13). Therefore, the direct beam radiation on a tilted surface, $I_{db, tilt}$ ($\text{W}\cdot\text{m}^{-2}$), is calculated using Equation (16):

$$I_{db, tilt} = I_{db, hor} \cdot R_b \quad \text{Eq. 16}$$

Here, $I_{db, hor}$ ($\text{W}\cdot\text{m}^{-2}$) shows direct beam radiation on a horizontal surface.

The sky diffuse radiation on a tilted surface, $I_{sd, tilt}$ ($\text{W}\cdot\text{m}^{-2}$), is calculated using Equation (17) [46]:

$$I_{sd, tilt} = I_{d, hor} \cdot \left(\frac{1 - A_i}{2} \cdot (1 + \cos \beta_{sur}) + A_i \cdot R_b \right) \quad \text{Eq. 17}$$

where $I_{d, hor}$ ($\text{W}\cdot\text{m}^{-2}$) is diffuse radiation on a horizontal surface, and A_i (–) is anisotropy index, given by Equation (18) [46]:

$$A_i = \frac{I_{bn}}{I_{on}} \quad \text{Eq. 18}$$

where I_{bn} ($\text{W}\cdot\text{m}^{-2}$) is beam radiation at normal incidence, and I_{on} ($\text{W}\cdot\text{m}^{-2}$) is extraterrestrial radiation at normal incidence.

Furthermore, the ground-reflected diffuse radiation on a tilted surface, $I_{gd, tilt}$ ($\text{W}\cdot\text{m}^{-2}$), is calculated using the ratio R_r (–), defined by Equation (19) [21]:

$$R_r = \frac{1 - \cos \beta_{sur}}{2} \cdot \rho_g \quad \text{Eq. 19}$$

where ρ_g (–) is the ground shortwave reflectance. Therefore, $I_{gd, tilt}$ ($\text{W}\cdot\text{m}^{-2}$) is obtained using Equation (20):

$$I_{gd, tilt} = I_{t, hor} \cdot R_r \quad \text{Eq. 20}$$

In the above equation, $I_{t, hor}$ ($\text{W}\cdot\text{m}^{-2}$) is the total radiation on a horizontal surface.

The total radiation incident on a tilted surface, $I_{t, tilt}$ ($\text{W}\cdot\text{m}^{-2}$), is the sum of the direct beam, sky diffuse, and ground-reflected diffuse radiation components, as calculated using Equation (21):

$$I_{t, tilt} = I_{db, tilt} + I_{sd, tilt} + I_{gd, tilt} \quad \text{Eq. 21}$$

Finally, the total solar shortwave radiation heat gain, $I_{t, tilt, srg}$ ($\text{W}\cdot\text{m}^{-2}$), is given by Equation (22):

$$I_{t, tilt, srg} = \alpha_{sur} \cdot (I_{db, tilt} + I_{sd, tilt} + I_{gd, tilt}) \quad \text{Eq. 22}$$

where α_{sur} (–) is the solar shortwave absorptance and subscript *srg* shows solar shortwave radiation heat gain.

2.2.3. Building energy modeling in CityBEM

CityBEM predicts transient building thermal loads to estimate total energy consumption, including HVAC, lighting, and equipment. Fig. 5 (a) illustrates the various heat transfer terms considered for a typical building in CityBEM. The arrow directions in Fig. 5(a) do not necessarily indicate heat flow direction. CityBEM currently assumes a single thermal zone for each building, with well-mixed indoor air. This enables the model to estimate average indoor conditions and thermal loads. Moreover, a constant air volume (CAV) HVAC system is simulated for heating

and cooling. Fig. 5(b) shows the thermal resistance model for conduction through building surfaces, while Fig. 5(c) displays various factors considered in glazing modeling.

Based on the schematic of the thermal-resistance model shown in Fig. 5(b) and 1D conduction through building surfaces is modeled using Equations (23)–(26) [42,43]:

$$C_i \cdot \frac{\Delta T_i}{\Delta t} = \frac{T_{i-1} - T_i}{R_{i-1} + R_i} + \frac{T_{i+1} - T_i}{R_{i-1} + R_i} + S_i \quad \text{Eq. 23}$$

$$C_i = \rho_i \cdot \Delta x_i \cdot C_{p,i} \quad \text{Eq. 24}$$

$$R_i = \frac{R_t}{2(N-1)} \quad \text{Eq. 25}$$

$$R_t = \frac{L}{k} \quad \text{Eq. 26}$$

where subscript *i* denotes spatial notation for the *i*th thermal node. The variables used in the above four equations are: C ($\text{J}\cdot\text{K}^{-1}$) for thermal capacity; T (K) for temperature; t (s) for time, ρ ($\text{kg}\cdot\text{m}^{-3}$) for density, Δx (m) for layer thickness, C_p ($\text{J}\cdot\text{kg}^{-1}\cdot\text{K}^{-1}$) for specific heat, R ($\text{m}^2\cdot\text{K}\cdot\text{W}^{-1}$) for conduction thermal resistance between the midpoint of a cell and its boundary, S ($\text{W}\cdot\text{m}^{-2}$) for the source term representing a heat source, N (–) for the total number of thermal nodes in a surface layer, including the boundaries, L (m) for entire layer thickness, k ($\text{W}\cdot\text{m}^{-1}\cdot\text{K}^{-1}$) for thermal conductivity, and R_t ($\text{m}^2\cdot\text{K}\cdot\text{W}^{-1}$) for the conduction thermal resistance of the total surface layer. The numerical method applied to solve the conduction heat transfer equations is discussed in detail in previous works [43].

For boundary nodes in contact with outdoor air, R_{sky} and R_{ground} are thermal resistances accounting for longwave radiation with the sky and ground, respectively, while $R_{air, out}$ is the thermal resistance which accounts for both longwave radiation and convection with outdoor air [42, 43]. Moreover, the source term for boundary nodes in contact with outdoor air, S_N ($\text{W}\cdot\text{m}^{-2}$) shown in Fig. 5(b), represents the total solar shortwave radiation heat gain by exterior surfaces ($I_{t, tilt, srg}$), given by Equation (22). Equation (27) is used to calculate the source term S_N , considering the solar shading impacts [21,22]:

$$S_{N, shd} = I_{t, tilt, srg, shd} = \alpha_{sur} \cdot (I_{db, tilt} \cdot f_{shading} + I_{sd, tilt} + I_{gd, tilt}) \quad \text{Eq. 27}$$

where $f_{shading}$ (–) is the shading factor introduced in Equation (8), and the subscript *shd* represents direct solar shading impacts.

For boundary nodes in contact with indoor air, $R_{air, in}$ and $R_{th, mass}$ represent the thermal resistances accounting for convection heat transfer between the interior side of surfaces and indoor air and longwave radiation between the interior surfaces and thermal mass, respectively [42,43]. A similar one-dimensional conduction model (Eqs. (23)–(26)) is used for the floor and thermal mass, but the floor's exterior surface is near the ground, while the thermal mass's exterior surface is near the indoor area.

Furthermore, Table A1 in the appendix summarizes the key mathematical models used in CityBEM to calculate total thermal load, including thermal heat gain/loss through building surfaces, glazing heat balance, solar energy flow through the glazing, infiltration thermal load, internal heat loads, as well as the calculation of indoor temperature, HVAC and total building energy consumption.

2.3. Urban building energy modeling with the enhanced CityBEM framework

This section discusses the enhanced CityBEM framework developed in this study for the modeling of detailed, scalable, and transient urban buildings, including solar shading interactions. Although existing UBEMs often simplify or omit shading effects because of complexity and performance concerns, the enhanced UBEM workflow introduced here

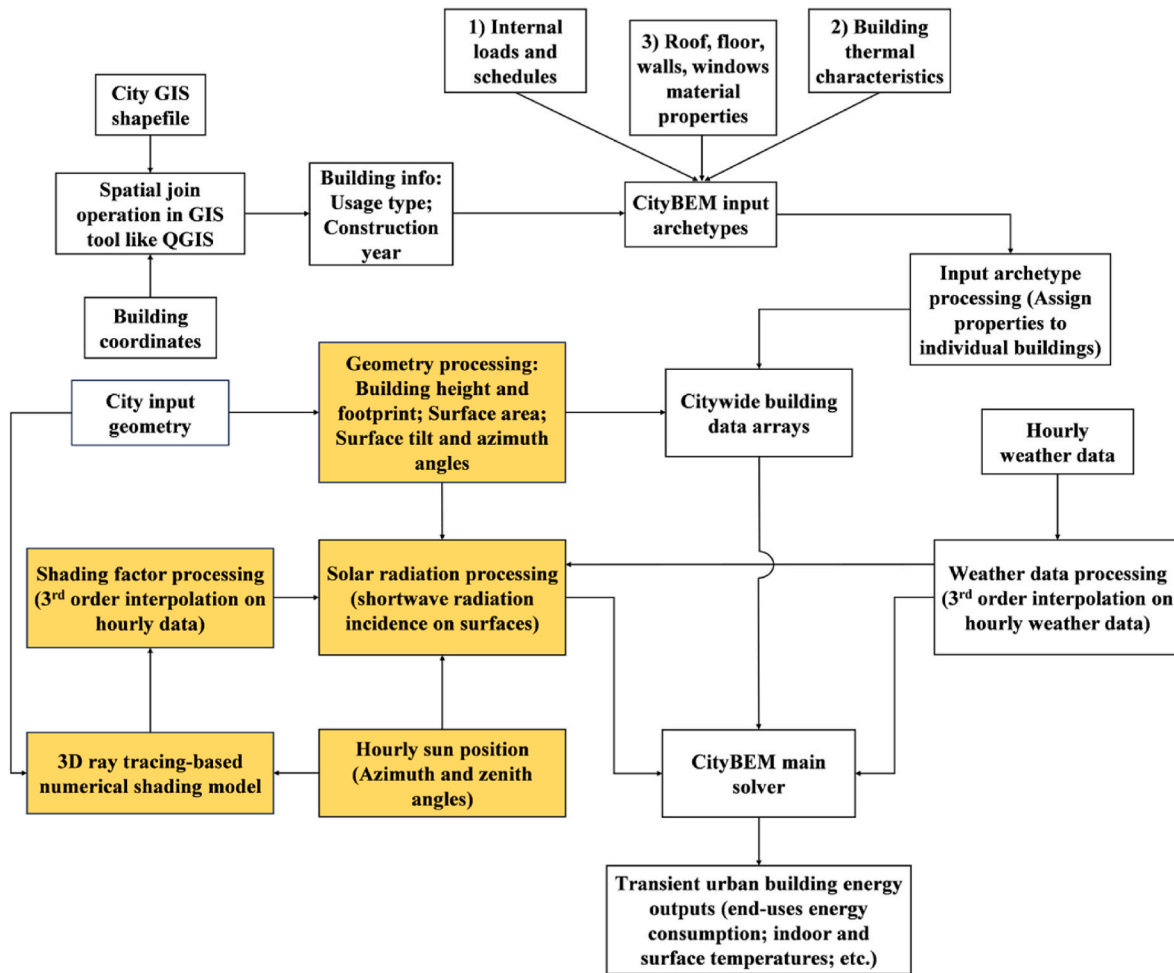


Fig. 6. Enhanced CityBEM framework by integrating the 3D ray tracing-based solar shading model (highlighted boxes indicate the new developments in this study).

allows for accurate computation of solar radiation and shading impacts with minimal computational overhead. Fig. 6 illustrates the new features added to the previous CityBEM workflow for urban building energy modeling [42].

The major UBE stages in the enhanced CityBEM framework are summarized below:

- i. Geospatial data integration – Assigning building metadata (e.g., location, year of construction, usage type) using GIS tools like QGIS [47].
- ii. Building property assignment from input archetypes – Allocating schedules, internal heat loads, envelope properties, and HVAC system characteristics on 38 usage types and 10 construction periods, using a 380-archetype library adapted from following references for North American cities: Quebec Construction Code and Safety [48], Ontario Building Code [49], ASHRAE Standard 90.1 [50], and the U.S. Department of Energy reference buildings [51].
- iii. 3D geometry processing – Extracting surface properties from urban geometry by supporting LOD 1 at present.
- iv. Weather interpolation – Aligning hourly weather input with the simulation time step.
- v. 3D ray tracing shading model – Calculating hourly shading factors based on urban 3D geometry and hourly sun position. The solar radiation calculation incorporates these factors.
- vi. Solar radiation processing – Calculate surface-level incident solar radiation accounting for direct, diffuse, and reflected components under shading.

- vii. Citywide data array generation – Package inputs for fast, transient simulation across all buildings.

It is worth noting that CityBEM's performance has been validated at both the city and individual building levels. In single-building studies, predictions are compared with hourly measured indoor temperature [42] and electricity demand [43]. Neighborhood level validation is carried out by comparing predictions against utility bills [52], whereas for a large portion of the city, predictions are compared against national benchmarks for energy intensity [42].

3. Model validation

3.1. Experimental validation of the solar shading model

This section compares the predictions of the current shading model with measurement data from Rocha et al. [53], who investigated three different configurations of a fin-like exterior shading structure connected to a vertical target surface. They measured the hourly sunlit fraction (or shading factor) on the target surface for different sun positions during the daylight hours. Fig. 7(a) illustrates the geometry and dimensions of one of these configurations, which we use for validation in this study. The experiment was conducted on May 2, 2015, in Curitiba, Brazil (25.52°S latitude, 49.17°W longitude), located in the southern hemisphere. The structure in the experiment is orientated northward, which is expected to receive the most direct solar radiation in the southern hemisphere. Further, solar azimuth and zenith angle data, which were used as inputs for the shading model, are presented in their

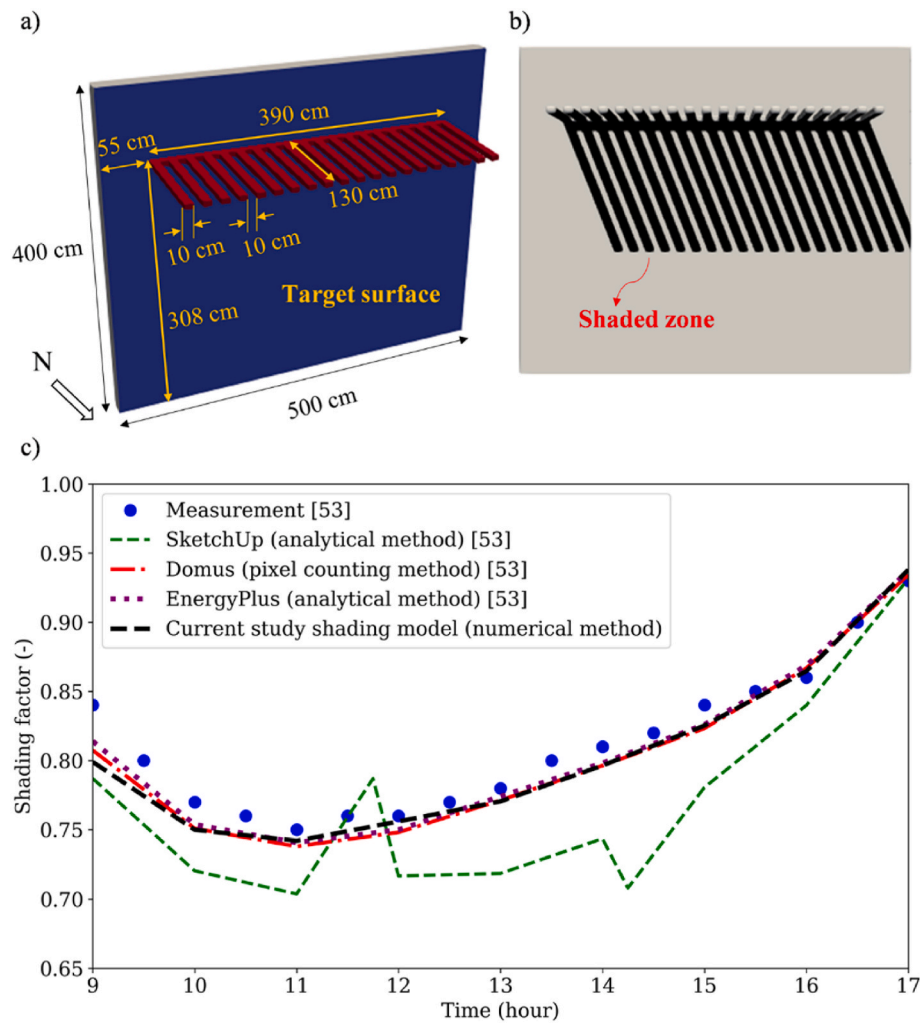


Fig. 7. Validation of the shading model with measurement data [53]: a) Input geometry; b) 3D view of the shading at 11 a.m. produced by the current shading model; c) Comparison of shading factors between measurement data, current model, and other tools.

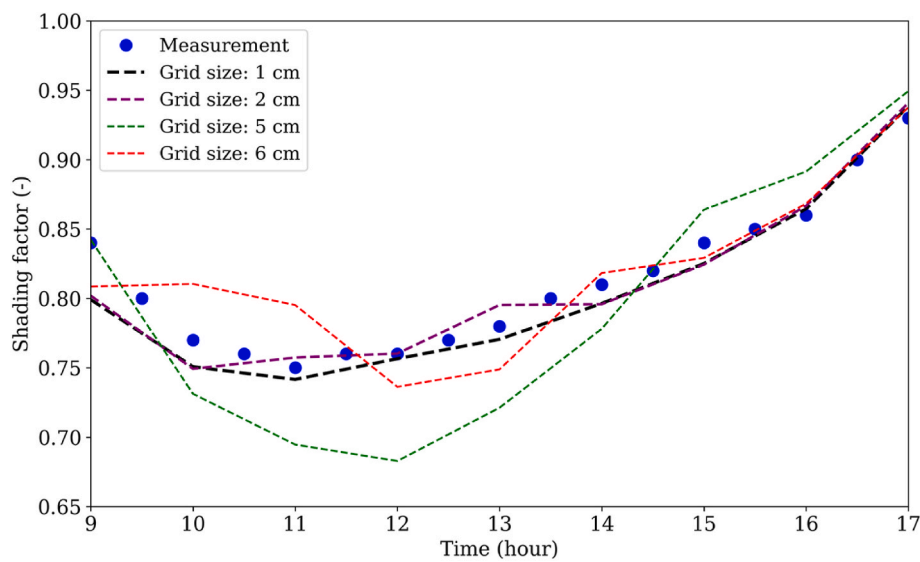


Fig. 8. The shading model predictions across mesh grid sizes.

Table 1

Performance evaluation of the shading model across grid sizes for the validation test case.

Grid size	Number of grid points	MaxAPE (%)	MAPE (%)	Test case run time (s)
6 cm	174,087	≈6	≈3	1
5 cm	289,952	≈10	≈5	4
2 cm	4,530,500	≈5	≈2	20
1 cm	36,244,000	≈5	≈2	160

study.

Fig. 7(b) shows a 3D view of the solar shading generated by the current shading model at 11 a.m., visualized using ParaView, a free and open-source data visualization tool [54]. It should be noted that the shading factor ($f_{shading}$) at each hour represents the fraction of the target vertical surface (Fig. 7(a)) exposed to direct sunlight and is unaffected by the shading of the exterior structure. In other words, the shading factor is calculated by dividing the area of the shaded zone (see Fig. 7(b)) by the entire area of the target surface (see Fig. 7(a)). Fig. 7(c) compares the shading factors calculated from the measurement data with those predicted by the current shading model, SketchUp's shading plug-in, Domus, and EnergyPlus. The maximum absolute percentage error (MaxAPE) between the measurement data and the results generated by the current shading model is approximately 5%, which is consistent with the accuracy of EnergyPlus (MaxAPE = 3%) and Domus (MaxAPE = 4%). In comparison, SketchUp's shading plug-in exhibits a higher MaxAPE of about 14%. According to these results, the numerical predictions derived from the current shading model are in good agreement with measurement data and trends observed in other shading models, confirming the model's reliability and accuracy.

Furthermore, the accuracy of the shading model's predictions is affected by the grid size, as shown in Fig. 8 and Table 1. Generally speaking, reducing grid size improves accuracy compared to measurement data. For the above test case, as seen in Table 1, the mean absolute percentage error (MAPE) decreases from nearly 5% at a 5 cm grid size to approximately 2% at a 1 cm grid size. However, finer grids can significantly increase simulation time, from 1 s to 160 s for the entire simulation of the test case, when reducing the grid size from 6 cm to 1 cm, due to the additional computational cost involved. It is also worth noting that the 6 cm grid yielded slightly better accuracy than the 5 cm grid. This may seem counterintuitive, but it is possible to experience such behavior when grid sizes are close, as minor differences in how grid cells are aligned with complex geometric features (e.g., shading edges or fine projections) can affect accuracy. In the validation test case shown in

Fig. 7, a grid size of 1 cm is selected, which corresponds to 36,244,000 grid points, giving a MaxAPE of 5%, which closely matches the measurement data. It should be noted that Fig. 8 and Table 1 show no significant accuracy improvement between 2 cm and 1 cm grid sizes, suggesting that a 2 cm grid size will suffice for practical applications of the above test case.

3.2. Verification of solar radiation model in CityBEM

In this section, EnergyPlus is used to verify the solar radiation data generated by CityBEM under solar shading impacts for a test case with four nearby buildings, as shown in Fig. 9. The comparison focuses on the south-facing surface of the building colored in blue (see Fig. 9), which is prone to shading effects from the adjacent buildings. Both models simulate solar radiation continuously over 48 h (January 1st and 2nd). Identical inputs and simulation time steps are used to ensure that meaningful comparisons can be made. EnergyPlus is chosen due to its widespread recognition in building energy modeling and its ability to provide detailed solar radiation data for individual building surfaces under solar shading impacts. For this comparison, the primary differences between the two models lie in their shading and solar radiation methodologies. EnergyPlus uses an analytical shading model, while

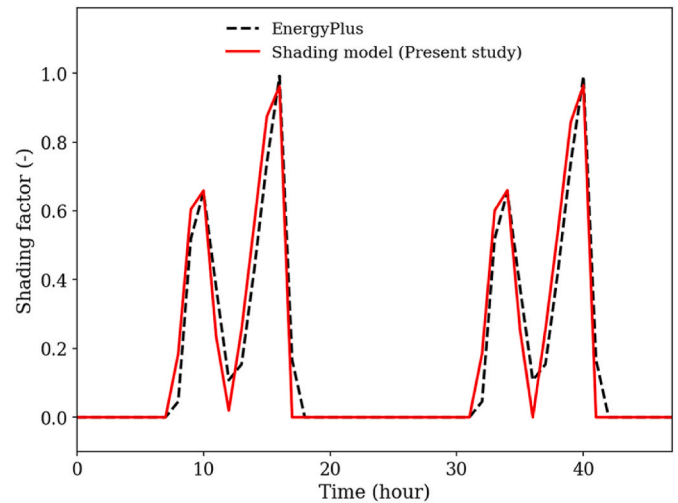


Fig. 10. Comparison of shading factors on the south façade of the blue building predicted by EnergyPlus and the present in this study.

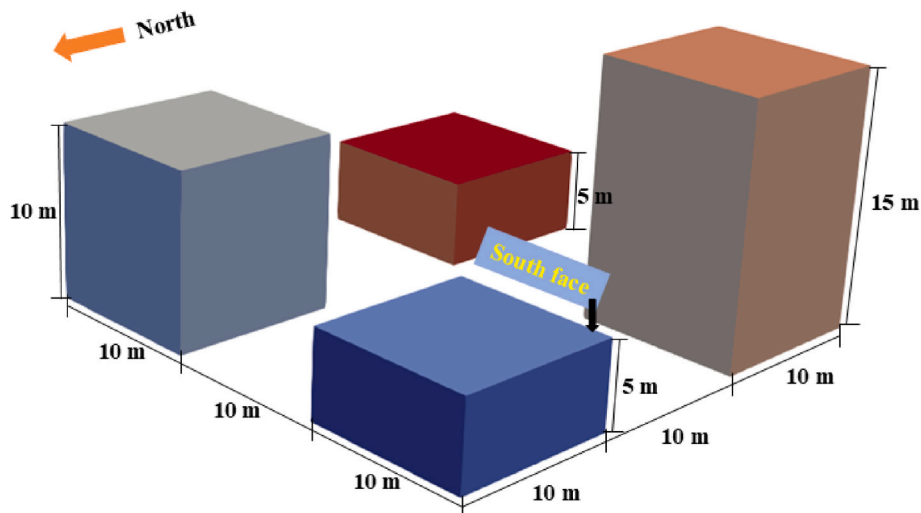


Fig. 9. Schematic of the test case used to compare the solar radiation data generated by CityBEM and EnergyPlus.

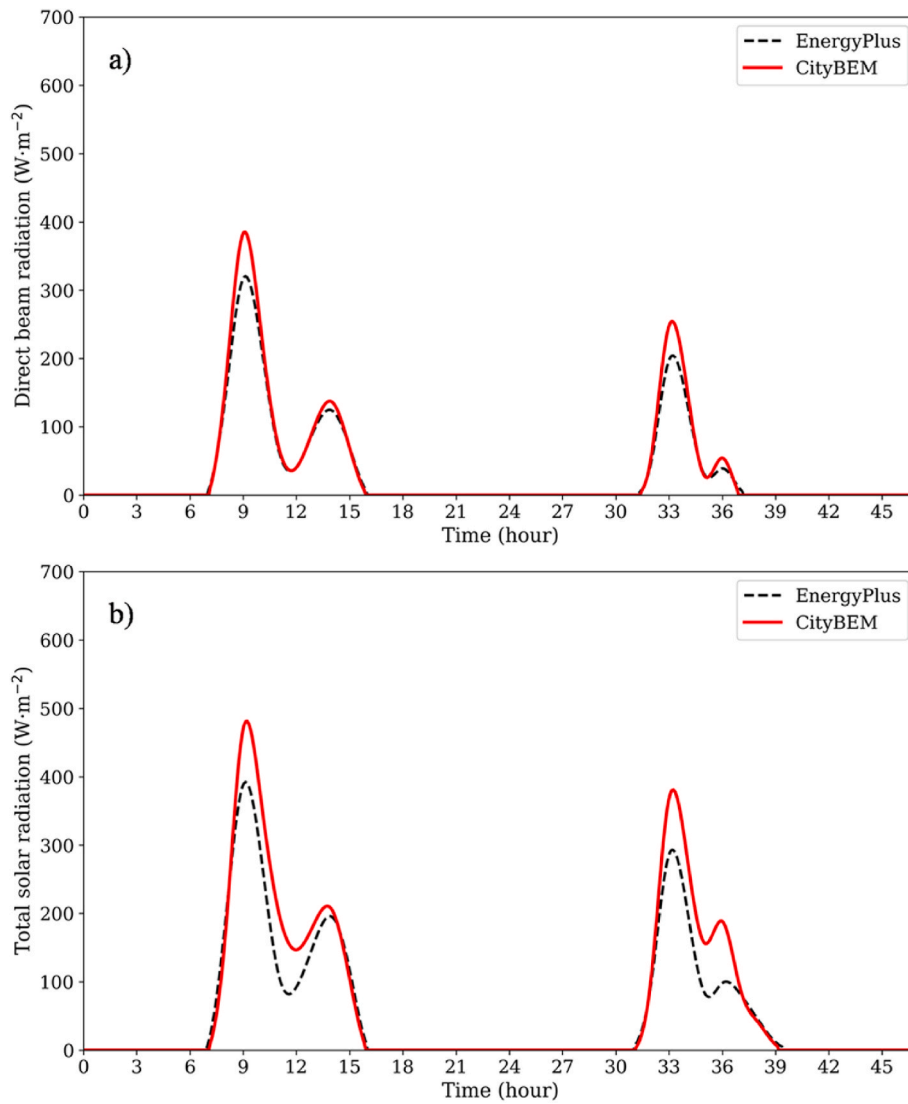


Fig. 11. Comparison of solar radiation terms calculated for the south-facing surface of the building (in blue) between EnergyPlus and CityBEM under solar shading impacts: (a) direct beam radiation; (b) total solar radiation.

CityBEM implements a ray tracing-based solar shading approach. The EnergyPlus engineering reference can be viewed [22] to better understand the radiation model used in EnergyPlus and for comparison with the CityBEM radiation model.

Fig. 10 illustrates the shading factor profile on the south façade of the building specified in Fig. 9, as predicted by EnergyPlus and the present shading model. As seen, the two models show good agreement, with similar shading factor trends during the two days.

Fig. 11(a) shows the comparison of direct beam radiation, and Fig. 11(b) shows the comparison of total solar radiation, both under solar shading impacts. The maximum absolute difference (MaxAD) for direct beam radiation is $66.23 \text{ W}\cdot\text{m}^{-2}$, and the mean absolute difference (MAD) is $5.02 \text{ W}\cdot\text{m}^{-2}$. The MaxAD is $91.86 \text{ W}\cdot\text{m}^{-2}$ for total radiation, and MAD is $15.52 \text{ W}\cdot\text{m}^{-2}$.

The discrepancies shown in Fig. 11 are consistent with those reported in previous comparisons of solar radiation predictions across various models—for example, between EnergyPlus, TRNSYS, and DOE-2, where discrepancies exceeding $150 \text{ W}\cdot\text{m}^{-2}$ have been reported [55]. These variations are well documented and are the result of differences in the underlying radiation models, which may use differing physical formulations, solar geometry resolutions, and shading algorithms. In particular, the models differ in how they approximate direct beam radiation,

account for sky diffuse and ground-reflected radiation, and represent nearby obstructions. Additionally, EnergyPlus and CityBEM use fundamentally different solar shading methodologies, which further contributes to the observed differences. It is also worth noting that the aim of this comparison is not to determine which method is better, but rather to demonstrate that CityBEM produces physically sensible and consistent solar radiation outputs when compared with other validated tools. Based on these comparisons, CityBEM's solar radiation estimates are reliable for subsequent building energy simulations.

4. Results and discussion

This section presents the results for evaluating the shading model across different complex scenarios as well as its use within the CityBEM framework for urban building energy modeling. The simulations presented are executed on a laptop equipped with an Intel® Core™ i7-10750H CPU @ 2.6 GHz and 16 GB of RAM.

4.1. Performance comparison of 3D shading model vs. analytical method

This section presents a quantitative runtime comparison between the 3D shading model in this study and the analytical shading feature in

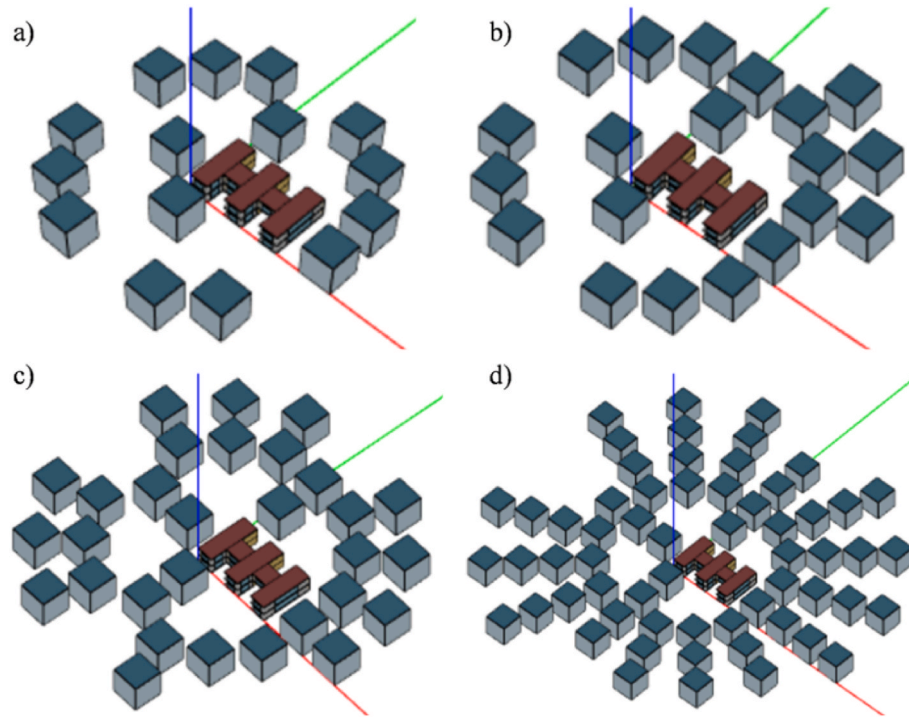


Fig. 12. Four hypothetical urban geometries used to study simulation runtime with increasing shading complexity: (a) 15, (b) 20, (c) 30, and (d) 60 surrounding shading buildings.

Table 2

Performance comparison of the 3D shading model in this study vs. EnergyPlus analytical shading feature for annual simulations.

Scenario (Fig. 12)	# of shading buildings	Annual sim. time (second/target surface)	Estimated full-scale sim. time (s, analytical model)	Full-scale sim. time (s, 3D model)
a	15	0.1	12	20
b	20	4.5	675	32
c	30	8.4	1646	66
d	60	34.5	11,937	100

EnergyPlus. Fig. 12 shows four hypothetical urban scenarios created to evaluate and compare simulation runtimes for varying levels of shading complexity.

As shown in Fig. 12, each scenario consists of one target building with 46 exterior surfaces exposed to sunlight, surrounded by 15, 20, 30, and 60 shading buildings, respectively. The runtime reported in EnergyPlus represents the computational time required to model the impact of all shading structures (gray cubic structures in Fig. 12) on these 46 target surfaces. Therefore, to estimate the total simulation time required for the full-scale scenario (i.e., treating every surface as target), the runtime per every target surface is calculated and scaled accordingly. This is necessary because the 3D model in this study computes shading interactions for every exterior surface in the domain, unlike EnergyPlus, which typically calculates shading only for surfaces of buildings with active thermal zones. Table 2 compares the computational time between the 3D shading model and the analytical shading method in EnergyPlus for annual simulations of the four hypothetical geometries shown in Fig. 12.

As shown in Table 2, the simulation time using the analytical method increases rapidly with the number of shading buildings, due to surface-by-surface shading calculation. It is evident that computational acceleration schemes are necessary when using the analytical method in complex urban geometries. By contrast, the 3D shading model maintains significantly lower simulation times, demonstrating its ability to model

large-scale urban shading calculations efficiently.

4.2. Shading modeling of complex geometries

The shading model can process a wide range of surfaces, including flat, curved, and objects with dense exterior surfaces nearby, such as trees. Since the model relies on ray tracing, it is particularly well suited to capturing shading effects in these complex geometries. Fig. 13(a) illustrates the geometry of a test case involving a tree and an object with a complex surface configuration. This test case has an area of approximately $25 \times 20 \text{ m}^2$ and a height of about 11 m. Fig. 13(b) shows 3D shading views of this test case under two different sun positions, corresponding to morning (left figure) and afternoon (right figure). There are over 11,000 surfaces comprising the tree and the nearby structure in this test case, making it a challenging shading modeling scenario. The presence of structures like trees can significantly complicate the shading calculations since most of the computational acceleration schemes might not be feasible in such instances [56].

Furthermore, it is worth noting that the shading impact from trees on low-rise buildings can be substantial, making modeling of this impact especially useful for UBEMs in neighborhoods with low-rise buildings, where tree shading can significantly influence thermal loads and energy consumption [57].

4.3. Shading modeling of large urban areas

This section describes a test case selected to show the model's ability to capture direct solar shading between buildings in large geometries without applying acceleration schemes. A real urban area with over 1300 buildings comprising 27,000 surfaces is modeled for this purpose, as shown in Fig. 14(a). This area measures approximately 3 km by 3 km, with buildings reaching 180 m in height. Fig. 14(b) and (c) show the 3D shading view of this large area without any simplifications made to building shapes or their exterior surfaces.

Here, a structured mesh with a resolution of 2 m is generated. In contrast to the small-scale validation test case presented in Section 3.1,

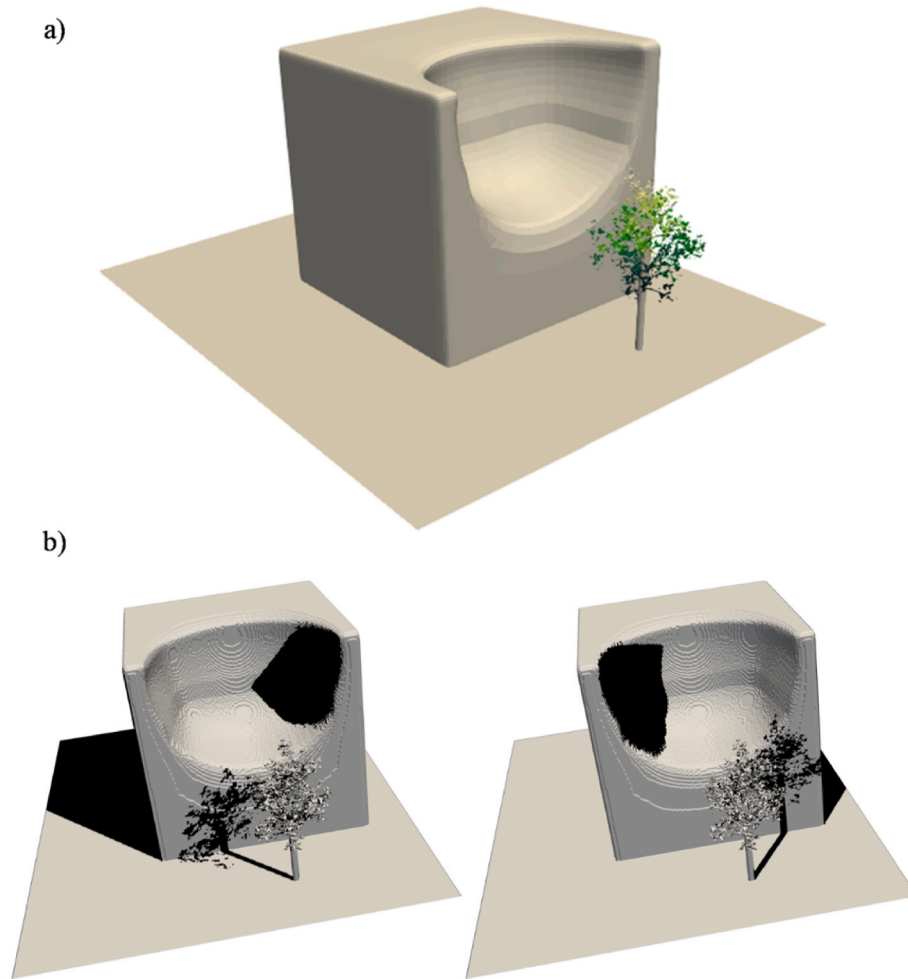


Fig. 13. a) The geometry of a tree next to a structure with a complex surface configuration; b) 3D shading views produced by the model at two different sun positions.

where finer grids (e.g., 1–5 cm) were necessary to capture shading patterns, the urban-scale simulation involves significantly larger buildings and surfaces, making a coarser resolution more appropriate. Using our experience and visual inspection across different resolutions, we have concluded that a 2 m grid size effectively captures the shading between buildings while maintaining reasonable computational requirements. In this downtown area, each grid cell is approximately 4 m^2 , which is considerably smaller than the typical façades of downtown buildings, making it an ideal choice for UBEM applications. A finer grid, however, may be necessary in neighborhoods with low-rise buildings or complex geometries. In the future, adaptive mesh refinement should be implemented based on various parameters, such as building height, neighborhood density, and geometric complexity.

Furthermore, an annual simulation of this test case with a 1 h time step is completed in approximately 120 min on a laptop equipped with an Intel® Core™ i7-10750H CPU @ 2.6 GHz and 16 GB of RAM. The authors acknowledge the importance of benchmarking this runtime against shading features in mainstream building energy models like EnergyPlus. However, the analytical shading model in EnergyPlus is designed for small-scale neighborhoods with simplified geometries. To apply it to a 3 km by 3 km real-world urban area like this test case, extensive simplifications (e.g., reducing surface counts) and custom computational acceleration schemes are necessary. Consequently, there will be no fair comparison when such modifications are made to EnergyPlus' standard workflow.

In order to provide a rough, yet intuitive reference for computational costs, we include results from Wang et al. [13], who evaluated the

performance of analytical-based methods for a neighborhood with 39 buildings. The hourly annual simulation of their test case took approximately 29 h using an analytical method. By utilizing computational acceleration schemes, this runtime was reduced to 145 min. Comparatively, the 3D ray tracing model in this study simulated a large test case with 1300 buildings in 120 min. This performance demonstrates the efficiency of the shading model in this study for large-scale, complex urban geometries.

Overall, it can be said that the practical applications presented in Section 4.1 to Section 4.3 highlight the shading model's robustness, particularly in handling intricate geometries and large-scale real urban environments without requiring preprocessing methods, which are often necessary for analytical methods. Moreover, the model consistently performs well under challenging shading conditions, providing evidence of its capability to simulate complex shading effects in a variety of simulation scenarios.

4.4. Building energy modeling of a real urban area using enhanced CityBEM workflow

Here we demonstrate the application of our detailed 3D ray tracing shading model to city-scale UBEM simulations, demonstrating a powerful alternative to conventional analytical methods. While the importance of solar shading in building energy simulation is well-established in the literature, the objective here is not to merely revisit that point, but to show how our 3D shading methodology can be seamlessly integrated into a complete UBEM workflow through a real-

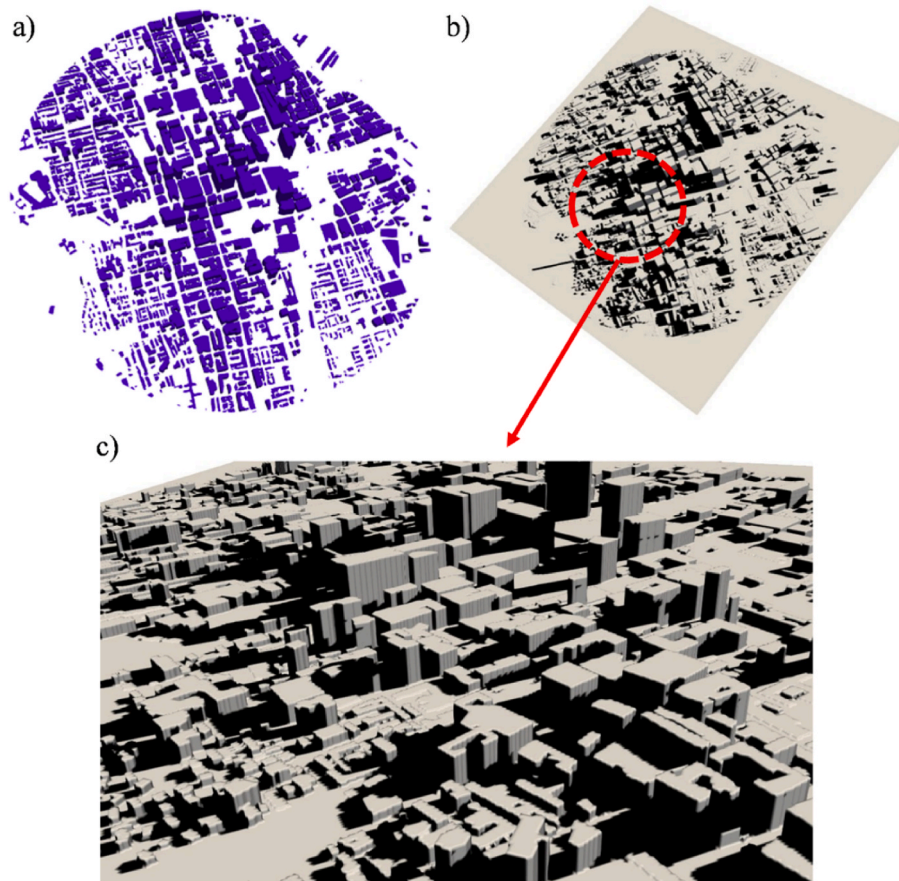


Fig. 14. 3D view of the shading generated by the current model for a large urban area with 1300 buildings and 27,000 shading surfaces.

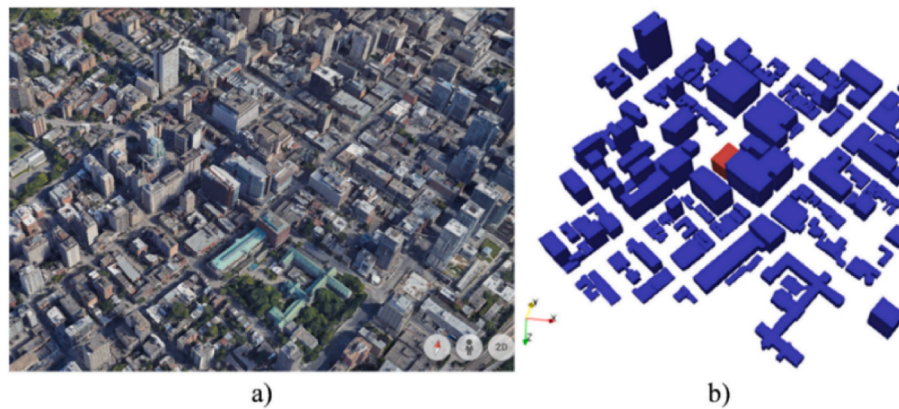


Fig. 15. Downtown Montréal test case with 255 buildings: a) the actual area in Google Earth, b) 3D geometries of buildings.

world, large-scale urban simulation. Previous shading studies have often underexplored the challenges of incorporating detailed shading models into UBEMs. In contrast, we presented a fully transparent methodology for implementing a 3D shading model within CityBEM and validated it through urban-scale simulations. A large downtown area is selected to fulfill the purpose of this demonstration. Downtown Montréal (Quebec, Canada) is chosen as the test case, as CityBEM's input archetypes are particularly well-suited to Canadian cities, including Montréal.

As shown in Fig. 15, the test case area spans approximately $0.7 \text{ km} \times 0.7 \text{ km}$ and contains 255 buildings of diverse usage types and heights (up to 113 m). The selected neighborhood includes a mix of low-, mid-, and high-rise buildings, making it suitable for testing the impact of shading in complex urban settings. As seen in Fig. 15(b)–a mid-rise educational

building (1600 m^2 , 45 m tall), is chosen to study the detailed impact of surrounding shading on its thermal load pattern.

The simulations for this test case are conducted for two representative weeks in 2023: January 1–7 and July 15–21, covering both cold, low-sun conditions and warm, high-sun conditions. The weather data used as inputs are shown in Fig. 16. These timeframes allow for assessing shading effects under these two distinct seasonal conditions.

In Fig. 17, shading patterns are shown in 3D from 11:00 to 14:00 in January as sample shading results. Solar shading factors are calculated hourly for each surface, and are interpolated at sub-hourly time steps in order to maintain the minute-level temporal resolution required by CityBEM simulations.

The thermal load variations for the selected educational building are

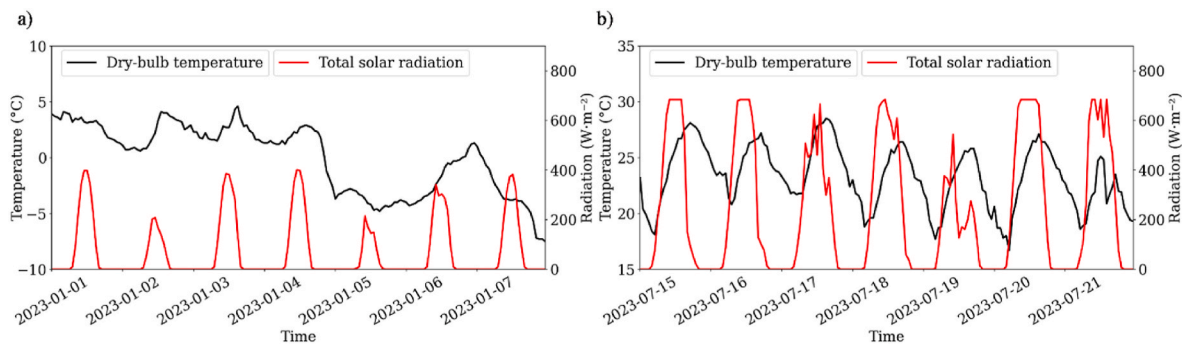


Fig. 16. Weather data used in CityBEM simulations: a) winter week, b) summer week.

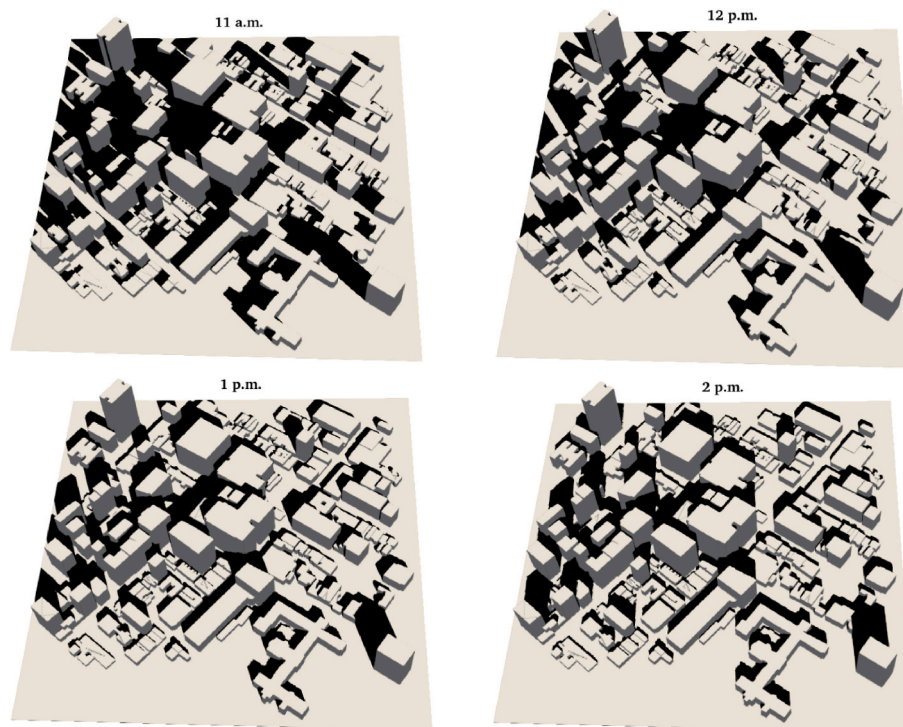


Fig. 17. 3D solar shading view for downtown Montréal test case from 11 a.m. to 2 p.m. in January, as simulated by the model in this study.

shown in Fig. 18. In winter week (Fig. 18(a)), the heating load is underestimated on sunny days when shading is ignored (e.g., days 1, 3, 4, 6, 7). Due to the absence of solar input, load differences at night tend to converge. Moreover, summer week (Fig. 18(b)) shows the opposite trend—cooling loads are underestimated without shading modeling, particularly on days when radiation levels are high.

Furthermore, Fig. 19 shows the cumulative daytime energy consumption of all buildings in the test case to maintain indoor thermal comfort. The energy use data are aggregated across all time steps from 6 a.m. to 6 p.m. (representing daytime energy consumption). During the winter, energy consumption is higher because of the cold temperatures in Montréal and the consequent high heating load. Furthermore, the building energy consumption remained below $10 \text{ kWh}\cdot\text{m}^{-2}$ for both weeks. By comparing cases with and without shading, we observe 13% on average underestimation of heating in the winter week and around 15% on average overestimation of cooling in the summer week when shading is ignored.

Fig. 20 also evaluates rooftop solar radiation overestimations caused by neglecting shading. Roughly 30% of buildings experience 10%–30% overestimation in both seasons. Over 34% of rooftops exceed 20% overestimation, with winter peaks reaching 50%, and some summer

cases exceeding 100%. Based on these results, large-scale simulations that do not account for shading often significantly overestimate the potential for solar rooftop power generation.

The results of this test case validates that the methodology is both scalable and computationally feasible for large, complex urban areas without requiring any geometry simplification or computational acceleration schemes. We hope our approach will guide future efforts to implement and extend shading models in UBEMs, supporting more accurate and scalable urban energy predictions and contributing to broader carbon neutrality goals.

4.5. Discussion on methodology, scalability, and future applications of this study

As this study relies on physics-based modeling approaches, its methodology is designed to be generalizable to different climate conditions and regions. In Section 3.1, the shading model is validated using experimental setups in Curitiba, Brazil, which has a hot, humid summer and a cool, dry winter. Additionally, CityBEM's performance has been demonstrated in previous studies across North American cities with cold climates [42]. These validations across diverse geographic and climatic

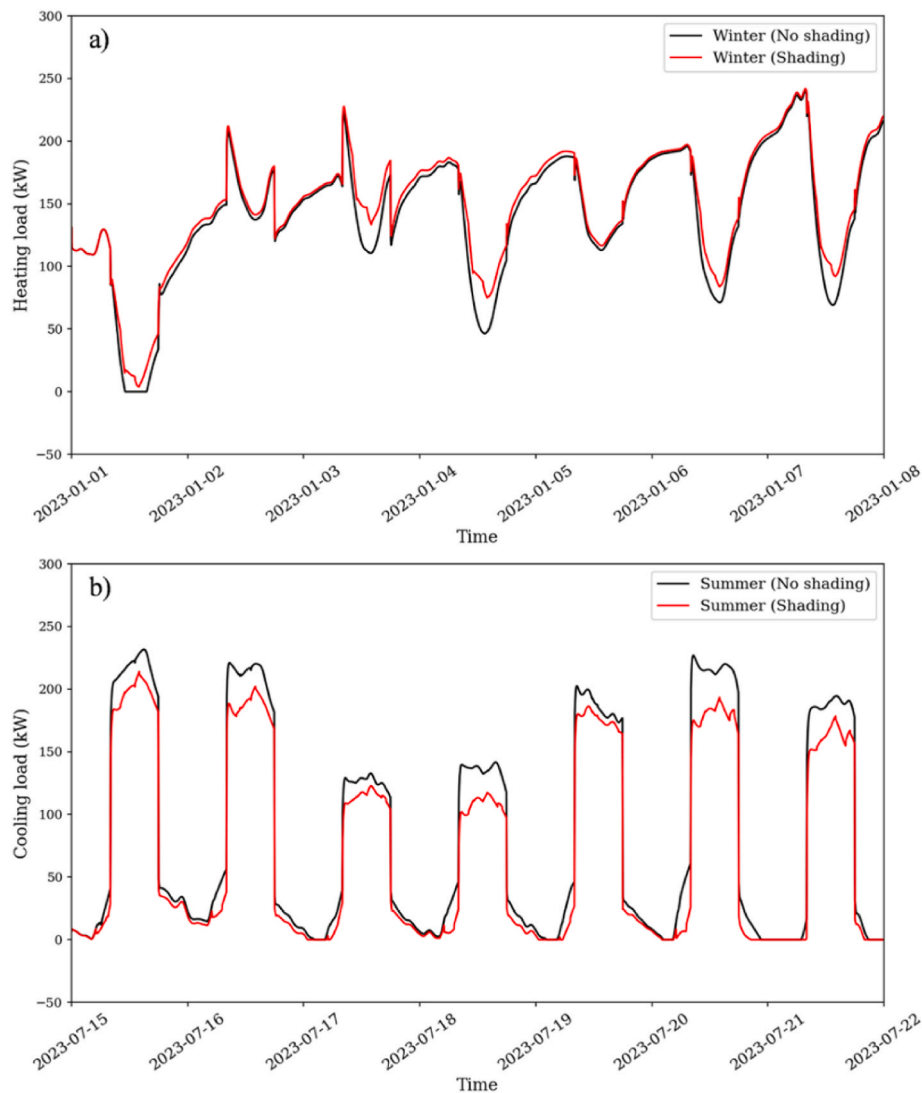


Fig. 18. Variation in thermal load of a sample building in downtown Montréal when the solar shading feature is turned on and off in CityBEM: a) winter week (heating load), b) summer week (cooling load).

contexts confirm that the models' accuracy remains robust and that local weather conditions do not inherently limit them. Furthermore, the Montréal region is selected as a test case in Section 4.4 because the current CityBEM's input archetypes are specifically designed for North American cities, particularly Canadian cities. The demonstration of the CityBEM's performance in other climates would be an excellent opportunity to enhance the CityBEM application in the future.

Understanding the impacts of solar shading on building energy performance can help designers and engineers optimize building shapes, locations, and orientations to enhance energy efficiency. The enhanced CityBEM framework can now support broader climate-responsive urban planning by guiding strategies (e.g., urban layout and design) to reduce heating, cooling, and lighting demands while optimizing on-site renewable energy production. Furthermore, coupling the model with urban microclimate tools enables a more comprehensive analysis of interventions such as green roofs, façade retrofits, and solar panels by exploring their effects on outdoor conditions while taking into account factors such as solar shading, building energy, and the urban microclimate [32]. By leveraging this capability, decision-makers can gain insight into balancing the implementation of energy efficiency measures, outdoor thermal comfort, and renewable energy adoption within complex urban environments.

4.6. Limitations and future work

This study has certain limitations that provide insight into future research directions. The shading model generates a structured grid across the entire domain, regardless of the input geometry size. While this works well for city neighborhoods with hundreds of buildings, it cannot be directly applied to larger areas, such as an entire city. In the next version, a geometry partitioning unit should be included to divide large geometries into smaller subdomains for faster processing. This method, well-known as the tiling method in computer programming [58], is commonly used for acceleration in large-scale simulations [29, 42]. Moreover, although the shading model is designed to be compatible with parallel processing methods, the current version uses serial processing. GPU parallel processing should be incorporated to meet the growing demand for faster urban simulations and to keep up with modern simulation tools.

In addition, future iterations of the shading model will incorporate an adaptive mesh resolution strategy based on the density of nearby exterior surfaces, such as trees or nearby buildings, to better capture complex shading patterns. In areas with substantial tree coverage, the mesh will automatically refine to a higher resolution to capture detailed shading effects, while coarser meshes will be applied in areas with fewer obstructions or during seasons with reduced foliage density (e.g.,

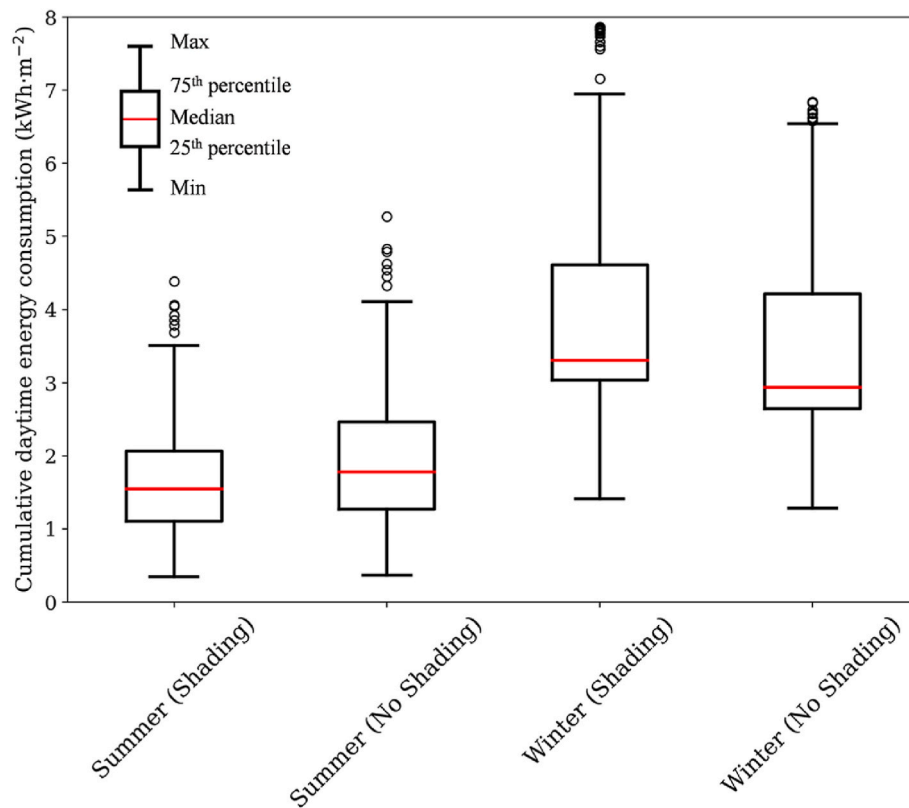


Fig. 19. Daytime energy consumption of downtown Montréal test case buildings with the shading model activation status in CityBEM.

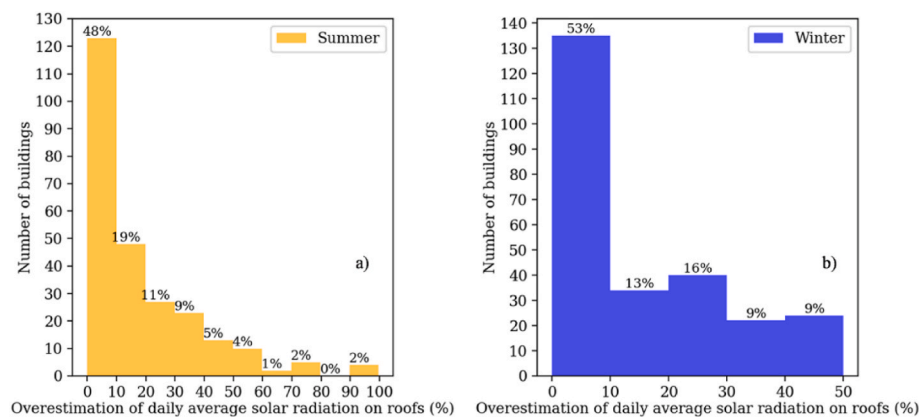


Fig. 20. Overestimation of daily average solar radiation on test case buildings' rooftops when shading is neglected: a) summer week, and b) winter week.

winter). Moreover, integrating species-specific growth patterns into the mesh generation process would improve the model's flexibility across different geographical regions, enabling more precise UBEM predictions.

5. Conclusions

A complex aspect of modeling solar radiation on urban building façades is the shading from adjacent structures. Analytical methods are commonly used to model shading effects in urban building energy models (UBEMs). Nevertheless, these methods do not provide conclusive results for urban-scale simulations due to the following limitations: (I) they require labor-intensive and difficult preprocessing methods, including input geometry simplification and computational acceleration schemes; (II) they cannot handle complex structures like curved buildings or dense nearby surfaces. We have developed a 3D ray tracing-based

solar shading model and integrated it into our in-house UBEM, CityBEM. The model is designed to process real-world, large-scale simulations without relying on preprocessing steps or acceleration schemes. Moreover, experimental data are used to validate the shading model's predictions.

Firstly, a ray tracing method is used to process diverse exterior surfaces (flat and curved), as well as dense shading elements such as trees, without the need to simplify geometry. The model captures solar shading interactions efficiently by dividing surfaces into small cells.

Secondly, the model adopts a 3D grid-based simulation approach that tracks sun rays across the entire urban scene simultaneously. This makes the model scalable for urban simulations and avoids the need for preprocessing and acceleration schemes, which are traditionally designed for analytical methods that calculate shading separately for each exterior surface. The model was tested on a variety of cases and successfully simulated shading for scenarios involving thousands of

surfaces.

Thirdly, the enhanced UBEM CityBEM follows a comprehensive methodology for solar radiation assessment that includes (I) determining the building surface properties (tilt, azimuth), (II) computing direct, diffuse, and ground-reflected radiation using physics-based models, and (III) taking into account shading effects from other buildings. This enhances the accuracy of the solar exposure calculation in UBEM.

In conclusion, the importance of modeling direct solar shading impacts on energy use predictions by UBEMs is underscored by our simulations for a large real case study in downtown Montréal, Canada. Neglecting shading led to a 13% average underestimation in winter heating demand and a 15% average overestimation in summer cooling demand. It also caused notable overestimations in rooftop solar radiation, with over a third of buildings exceeding 20%. In summary, our methodology improves UBEM energy predictions and supports better urban planning by accounting for direct solar shading.

Future work will facilitate scalable simulations for very large urban test cases by integrating additional features such as geometry partitioning, GPU acceleration, and automated inputs.

CRedit authorship contribution statement

Saeed Rayegan: Writing – review & editing, Writing – original draft, Visualization, Validation, Software, Methodology, Investigation, Formal analysis, Data curation, Conceptualization. **Mohammad Morteza-zadeh:** Writing – review & editing, Writing – original draft, Validation,

Software, Methodology, Investigation, Formal analysis. **Dongxue Zhan:** Writing – original draft, Visualization, Validation, Investigation. **Ali Katal:** Writing – review & editing, Validation, Methodology, Investigation. **Liangzhu Leon Wang:** Writing – review & editing, Supervision, Resources, Project administration, Methodology, Funding acquisition, Conceptualization. **Radu Zmeureanu:** Writing – review & editing, Supervision, Resources, Conceptualization. **Ursula Eicker:** Writing – review & editing. **Saeed Ranjbar:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Acknowledgments

The research is supported by the Axelys project “AXE- 0101 –An urban digital twin platform for greenhouse gas emission mitigation in Quebec” [#AXE-0101-UCON], the Contribution Grant “Forecasting indoor heatwaves and building responses against summertime heatwaves in Canadian cities” [#GCXE23S006] from Environment and Climate Change Canada (ECCC), the Natural Sciences and Engineering Research Council (NSERC) of Canada through the Discovery Grants Program [#RGPIN-2018-06734] and the Advancing Climate Change Science in Canada Program [#ACCPJ 535986-18].

Appendix A

Table A1

Summary of CityBEM mathematical formulation for modeling building thermal load and total power consumption (the methodology below can be found in detail in the following references: Rayegan et al. [42], and Katal et al. [43,52]).

Equations, parameters, subscripts, and information	Equation number
Envelope glazing heat balance (see Fig. 5 for the schematic of the glazing model):	(1) to (4)
$E_{out} \cdot \epsilon_{glz,out} - h_{rad,out} \cdot \theta_{out} + \frac{k_{glz}}{L_{glz}} \cdot (\theta_{in} - \theta_{out}) + h_{conv,out} \cdot (T_{air,out} - \theta_{out}) + S_{out} = 0$ $E_{in} \cdot \epsilon_{glz,in} - h_{rad,in} \cdot \theta_{in} + \frac{k_{glz}}{L_{glz}} \cdot (\theta_{out} - \theta_{in}) + h_{conv,in} \cdot (T_{air,in} - \theta_{in}) + S_{in} = 0$ $S_{out} = S_{in} = \frac{1}{2} \alpha_{glz} \cdot (I_{db,tilt} + I_{sd,tilt} + I_{gd,tilt})$ $S_{out,shd} = S_{in,shd} = \frac{1}{2} \alpha_{glz} \cdot (I_{db,tilt} \cdot f_{shading} + I_{sd,tilt} + I_{gd,tilt})$	
Variables:	
E ($W \cdot m^{-2}$): longwave radiation, ϵ (–): longwave emissivity, h ($W \cdot m^{-2} \cdot K^{-1}$): heat transfer coefficient, k ($W \cdot m^{-1} \cdot K^{-1}$): thermal conductivity, L (m): thickness, T (K): temperature, θ : glazing thermal node temperature (K), S ($W \cdot m^{-2}$): source term representing shortwave radiation heat gain, α (–): shortwave absorptivity, I_{db} ($W \cdot m^{-2}$): direct beam radiation, I_{sd} ($W \cdot m^{-2}$): sky diffuse radiation, I_{gd} ($W \cdot m^{-2}$): ground-reflected diffuse radiation, $f_{shading}$ (–): direct solar shading factor (–)	
Subscripts: <i>in</i> : indoor side, <i>out</i> : outdoor side, <i>air,in</i> : indoor air, <i>air,out</i> : outdoor air, <i>glz</i> : glazing, <i>rad</i> : longwave radiation interaction, <i>conv</i> : convection heat transfer, <i>tilt</i> : tilted surface, <i>shd</i> : direct solar shading impacts	
Information:	
It is assumed that there is no heat storage within the glazing layer due to its small thickness [22,43]. It is assumed that solar shortwave radiation is equally distributed on both sides of the glazing [22,59]. References [43,59] provide further details on the terms in the equations above and the numerical method used to solve this system.	
Solar energy flow through the building glazing, $q_{solar,glz}$ ($W \cdot m^{-2}$) (see Fig. 5 for the schematic of the energy flow through glazing): $q_{solar,glz} = SHGC_{glz} \cdot (I_{db,tilt} \cdot f_{shading} + I_{sd,tilt} + I_{gd,tilt})$	(5) & (6)
$q_{solar,glz,shd} = SHGC_{glz} \cdot (I_{db,tilt} + I_{sd,tilt} + I_{gd,tilt})$	
Variables:	
$SHGC$ (–): represents the solar heat gain coefficient	
Information: $SHGC_{glz}$ measures the total solar heat gain through glazing, including the directly transmitted solar radiation and the solar radiation absorbed by the glazing that is re-radiated as heat into the interior side.	
The total heat load of a building, Q_t (W):	(7)
$Q_t = Q_{sur} + Q_{inf} + Q_{internal}$	
Variables:	
Q_{sur} (W): building zone total heat gain/loss through building surfaces, Q_{inf} (W): infiltration thermal load, $Q_{internal}$ (W): internal heat loads	
Subscripts: <i>t</i> : total, <i>sur</i> : building surface, <i>inf</i> : infiltration	

(continued on next page)

Table A1 (continued)

Equations, parameters, subscripts, and information	Equation number
Building zone total heat gain/loss through exterior surfaces, Q_{sur} (W): $Q_{sur} = \sum_{i=1}^{N_i} A_{sur,i} \cdot h_{conv,sur,in,i} \cdot (T_{sur,in,i} - T_{air,in})$ Variables: N (–): surface number in contact with indoor air, A (m^2): cross-sectional area Subscripts: sur, in: interior building surface in contact with indoor air, i: surface counter Information: The mathematics of calculating $h_{conv,sur,in}$ is given in Ref. [43].	(8)
Infiltration thermal load, Q_{inf} (W): $Q_{inf} = F_{inf} \cdot \rho_{air,inf} \cdot C_p \cdot (T_{air,inf} - T_{air,in})$ $F_{inf} = \frac{V_b \cdot ACH}{3600}$ Variables: F_{inf} ($m^3 \cdot s^{-1}$): air infiltration rate, ρ ($kg \cdot m^{-3}$): density, C_p ($J \cdot kg^{-1} \cdot K^{-1}$): specific heat, V (m^3): volume, ACH (h^{-1}): air change per hour Subscripts: air, inf: infiltrated air, b: building Information: The equations used to calculate ACH can be found in Ref. [52].	(9) & (10)
Internal load of a building, $Q_{internal}$ (W): $Q_{internal} = Q_{occupants} + Q_{lighting} + Q_{equipment}$ Variables: $Q_{occupants}$ (W): occupants load, $Q_{lighting}$ (W): lighting load, $Q_{equipment}$ (W): equipment load Information: The three internal load components are calculated based on the schedules and loads for different building usage types, tabulated in previous studies [43,52].	(11)
Indoor air temperature, $T_{air,in}$ (K): $\rho_{air,in} \cdot V_b \cdot C_p \cdot \frac{\Delta T_{air,in}}{\Delta t} = Q_t + \dot{M}_{HVAC} \cdot C_p \cdot (T_{sup} - T_{air,in})$ Variables: $\Delta T_{air,in}$ (K): changes in indoor temperature over the time step Δt (s), $\dot{M}_{HVAC}$ ($kg \cdot s^{-1}$): HVAC system air flow rate Subscripts: sup: HVAC system supply air Information: Details about $\dot{M}_{HVAC}$ and T_{sup} can be found in the following references [43,59].	(12)
HVAC and building total power consumption, PC_{HVAC} and $PC_{b,t}$ (W): $PC_{HVAC,h} = \frac{Q_t}{\eta_h} + PC_{fan}$ $PC_{HVAC,c} = \frac{Q_t}{COP} + PC_{fan}$ $PC_{b,t} = PC_{HVAC} + PC_{lighting} + PC_{equipment}$ Variables: PC: Power consumption (W), η_h (–): heating efficiency, COP (–): cooling coefficient of performance Subscripts: $HVAC, h$: HVAC system heating, $HVAC, c$: HVAC system cooling, b, t: Building total Information: The mathematical description of PC_{fan} can be found in Ref. [59].	(13) to (15)

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